

Documentation

Electronics

BlueVision NG Automation Suite

Version 1.23

User Documentation

E532439/01E



Power. Passion. Partnership.



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1 Introduction

This user documentation describes the use and – as far as relevant for the user – the functionality of the BlueVision New Generation Automation Suite software.

The BlueVision New Generation Automation Suite is the software tool for commissioning and service of the following automation systems:

- ECS-6 (1163-04)
- BlueVision New Generation
- GenoLine New Generation

The main functions of the software tool are:

- Monitoring
- Stimulation
- Recording
- Dialog
- Configurator
- Data Retrieval

Note: The configurator functionality is not described in this document.

The access to certain program functions and SysIDs are only granted by the corresponding permissions (software licence). A licence for the corresponding automation system can only be granted after successful participation in a training course.

Always ensure that the latest version is used (see chapter 5 *Installation*).

MTU Friedrichshafen is constantly further developing their products. Please understand that this may result in deviations of the user documentation from the product.

This user documentation complies with BlueVision New Generation Automation Suite version V1.23.

2 Abbreviations

IP	Internet protocol
LAN	Local Area Network
MCS	Monitoring and Control System
MD	Missing Data
PG	Project Group
PLC	Programmable Logic Controller
RCS	Remote Control System
SCU	Systembus Coppling Unit
SD	Sensor Defect
SPU	Systembus Processing Unit
SysID	System Identifier

3 System Requirements

Processor:	Dual-Core with 2-GHz
Memory:	2GB RAM
Screen:	1024*768 (Minimum) / 1280*1024
Network:	Ethernet 100 Mbit/s
Operating system:	Microsoft Windows XP SP3 or Windows 7
Java Version:	5 (Minimum)

Note: The user must have the possibility to change the IP address of the network interface.

4 Preparation

For establishing a connection to the automation system, the service laptop intended for this task must be connected with the systembus by network cable (patch cable). The configuration of the network interface must comply with the address range of the systembus devices.

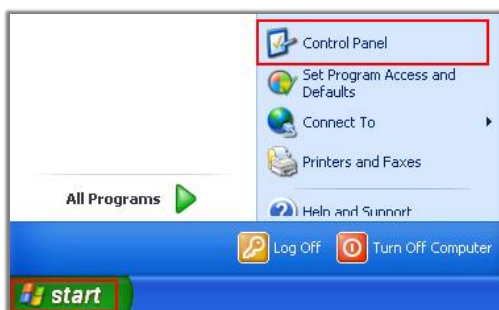
The service laptop must have the following network settings:

IP address:	10.0.240.[241 – 245]
Subnet mask:	255.0.0.0

4.1 Network Settings in Windows XP

Note: You need administrator rights to be able to make the following settings.

Open the **Control Panel**.



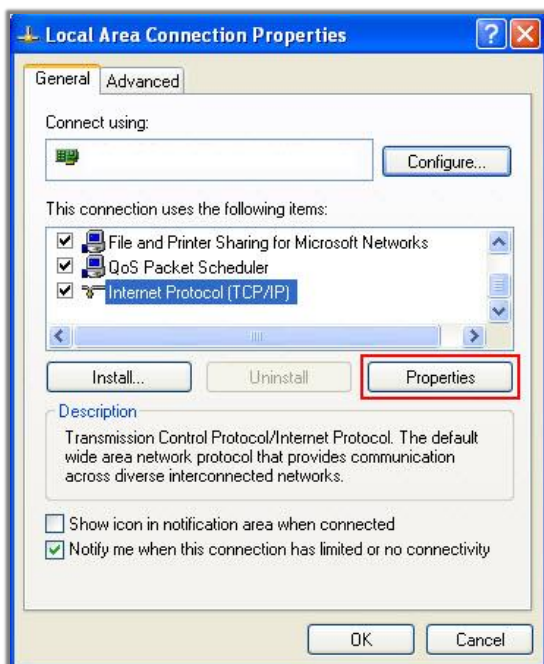
You will find the **Network Connections** under Control panel.



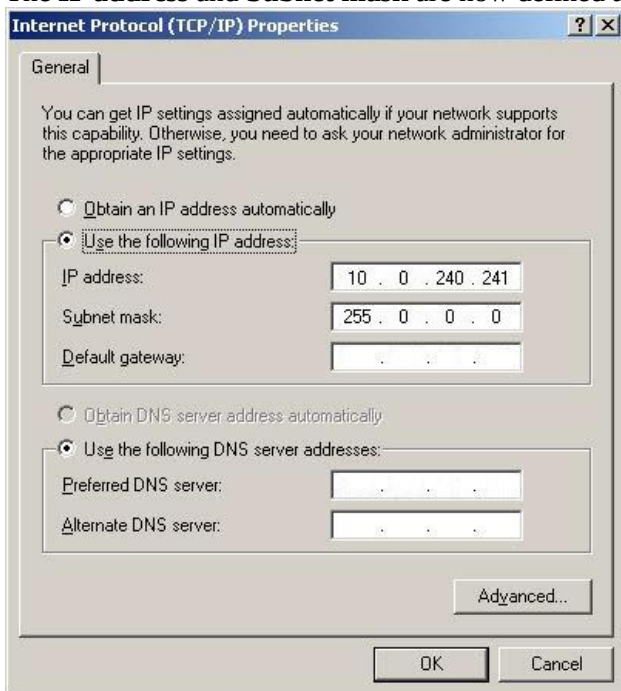
A screen opens displaying the network connections available on the service laptop.
Select the **Network Connection** which is connected with the systembus.
Right-click on the appropriate **LAN Connection** to open the context menu within the **Properties** menu item.



The **Properties** of the network connection contains different information and elements.
Mark the option **Internet protocol (TCP/IP)** and click on the Properties button.



The **IP address** and **Subnet mask** are now defined as shown in the following:



Note: The fields for **Default gateway** and **DNS server** are left empty.

Confirm with **OK** and click on the **extended** tab in the properties menu for the LAN connection in question.



Remove the checkmark next to the **Internet connection firewall** and confirm with **OK**.

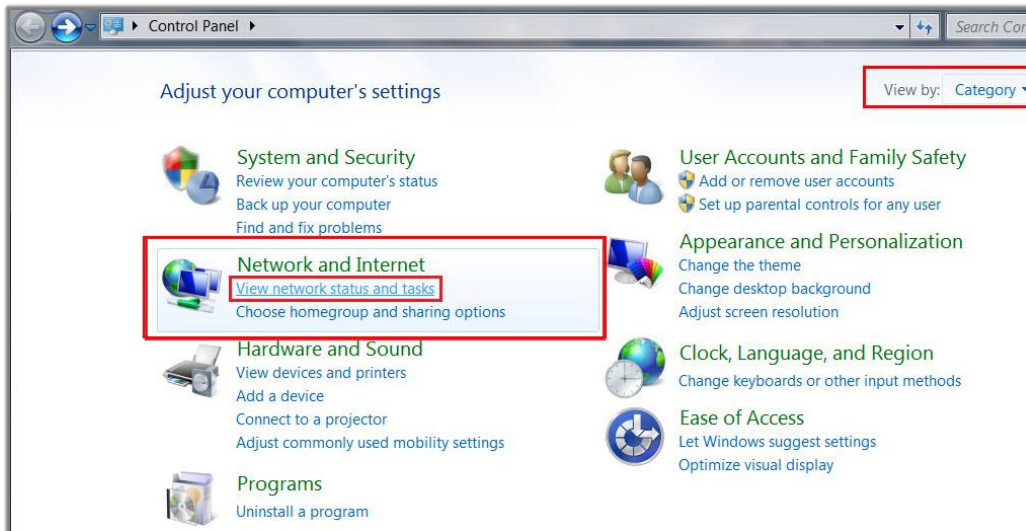
4.2 Network Settings in Windows 7

Note: You need administrator rights to be able to make the following settings.

Open the **Control Panel**.

Switch the display in the control panel to **Category**.

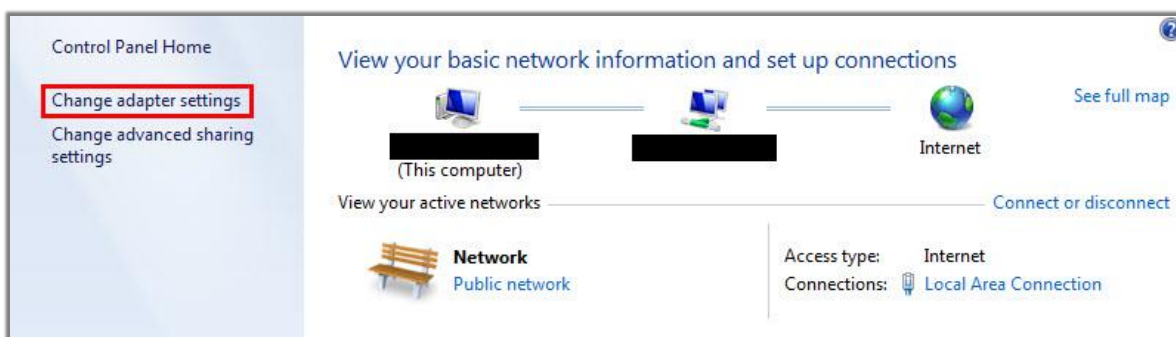
Next select the sub item **View network status and tasks** in the chapter **Network and Internet**.



The **Network and Sharing Center** opens.

There is a menu with different options on the left side.

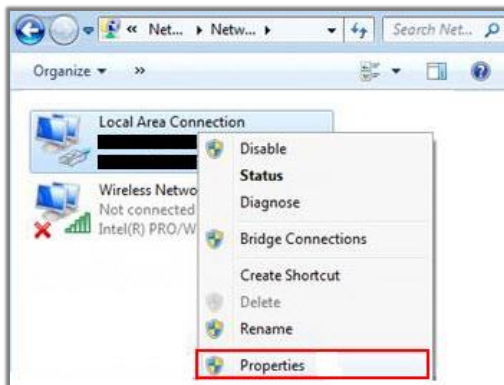
Click on the sub item **Change adapter settings**.



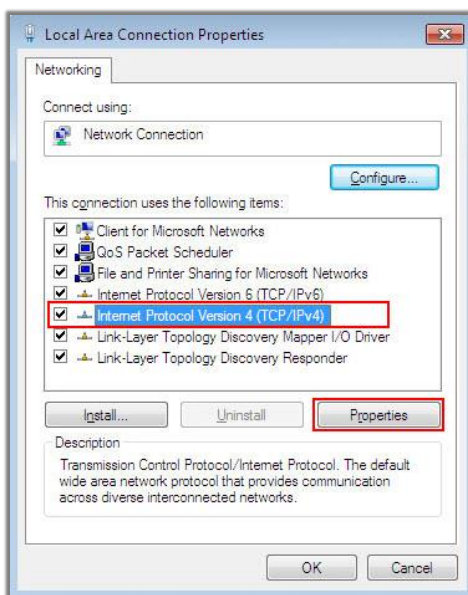
A screen with the network connections on the service laptop opens.

Select the **Network Connection** which is connected with the systembus.

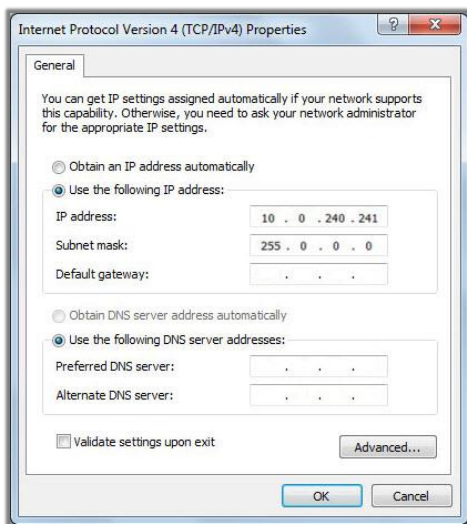
Right-click on the appropriate **LAN Connection** to open the context menu with the **Properties** menu item.



The **Properties** of the network connection contains different information and elements.
Mark the option **Internet Protocol Version 4 (TCP/IPv4)** and click on the Properties button.

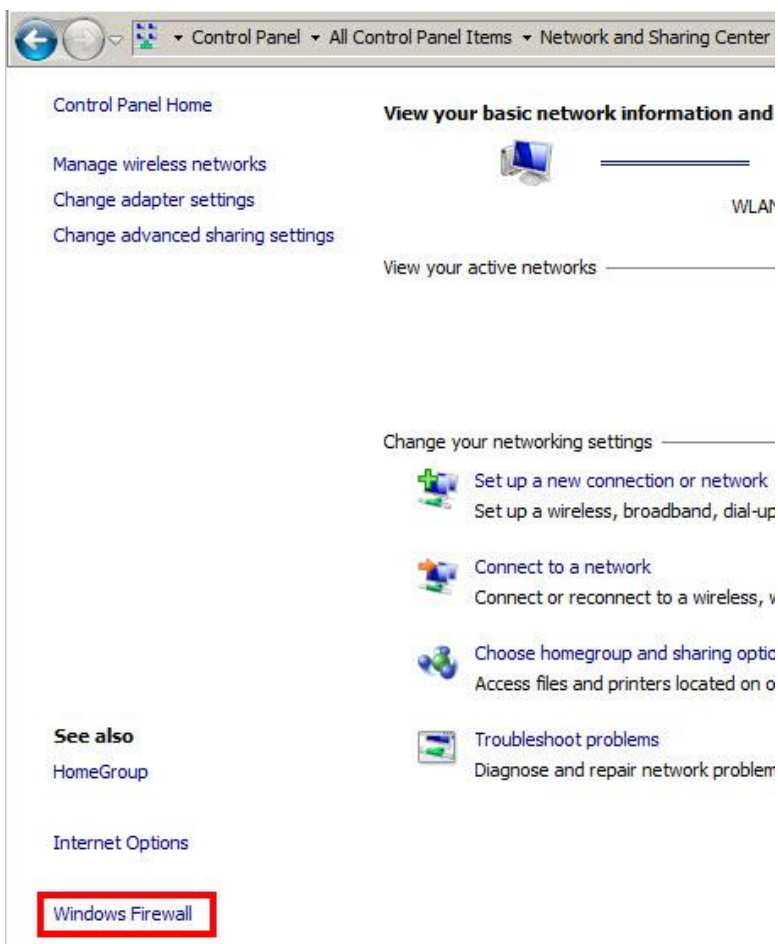


The **IP address** and **Subnet mask** are now defined as shown in the following:

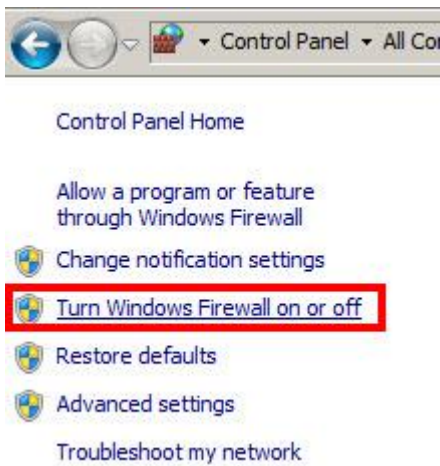


Note: The fields for **Default gateway** and **DNS server** are left empty.

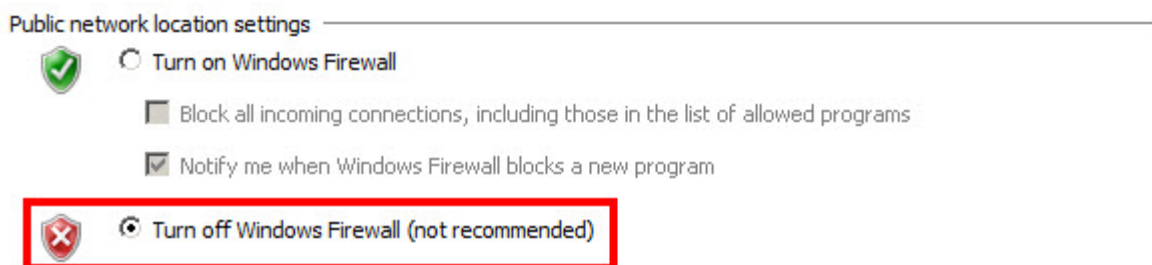
Confirm with OK and then re-open the **Network and Sharing Center**.
Deselect the **Windows Firewall** menu item in the lower left section



Deselect the **Turn Windows Firewall on or off** tab in the left section



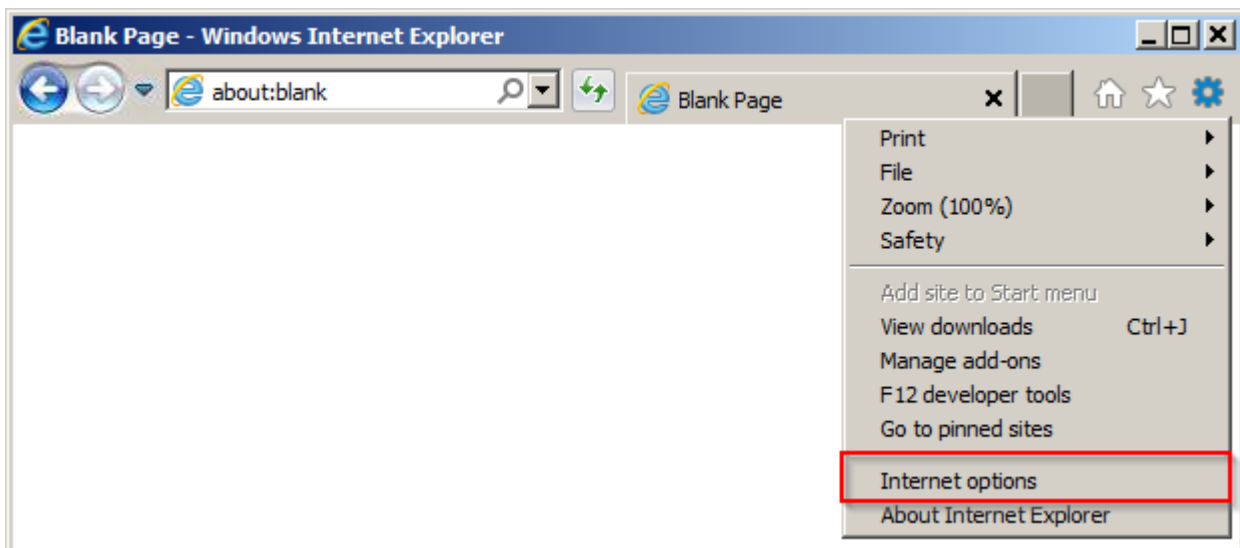
Deactivate the Firewall in the local settings for the public network by selecting the corresponding radio button and confirm with **OK**



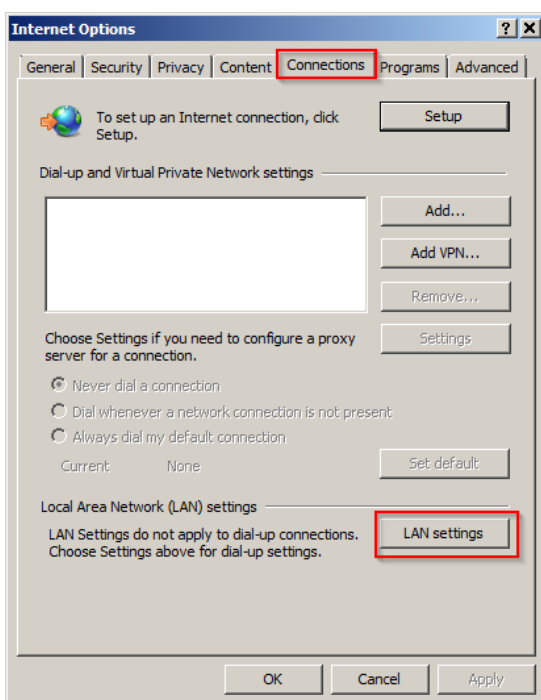
4.3 Proxy Server Settings (Microsoft Internet Explorer)

In case the website of an SPU node cannot be called you can check your Proxy server settings in Microsoft Internet Explorer as follows.

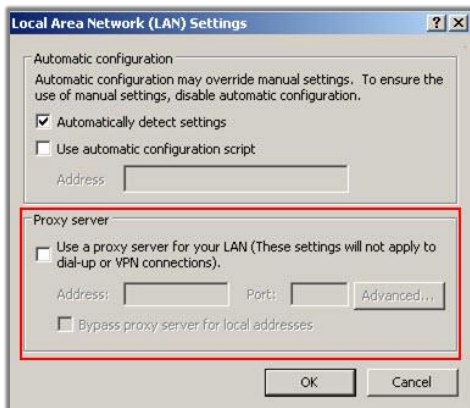
Open the Internet Explorer browser.
Select the **Internet options** in the **Extras** menu item.



The **Internet Options** opens.
Select the **Connections** tab and click on the **LAN Settings** button.



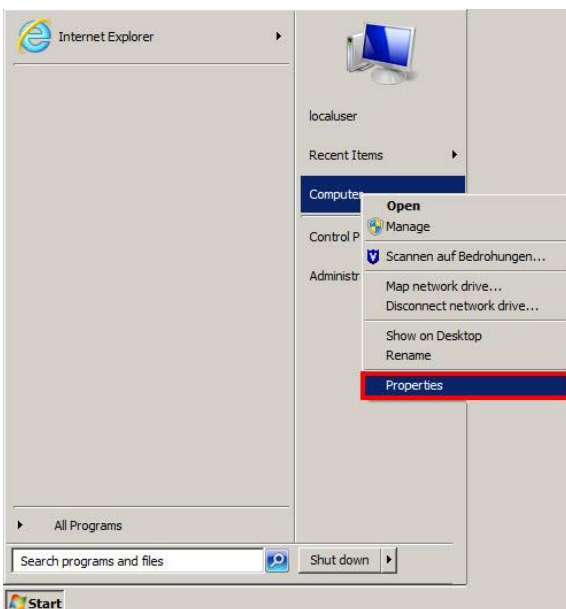
No **Proxy server** may be active in the **LAN Settings**.
Check whether the field is deactivated, and if necessary, remove the checkmark.



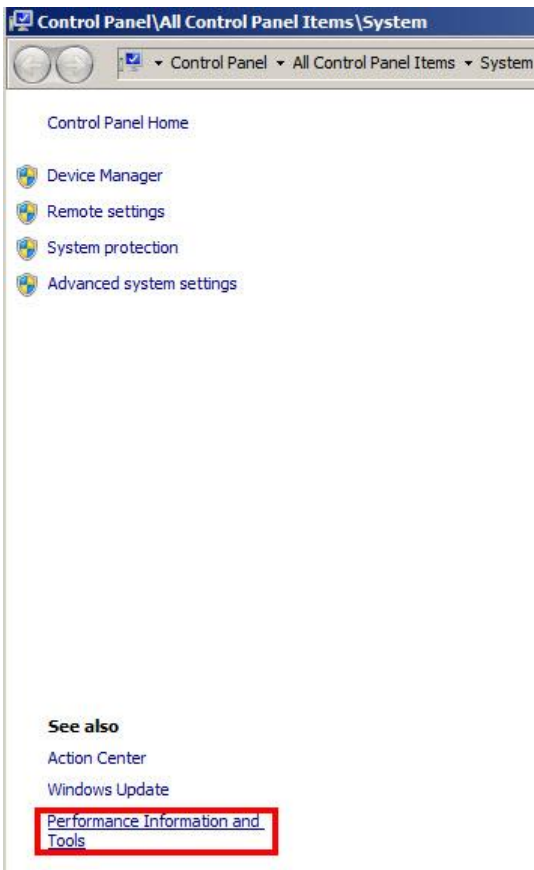
4.4 Visual Effects in Windows 7

In case the screen of the service-laptop switching to dark/black check the “Visual Effects”-settings in Windows as follows.

Start menu -> right-click on computer opens a Context menu -> Properties.



Open “**Performance Information and Tools**” at the bottom left.



Open “Adjust visual effects”

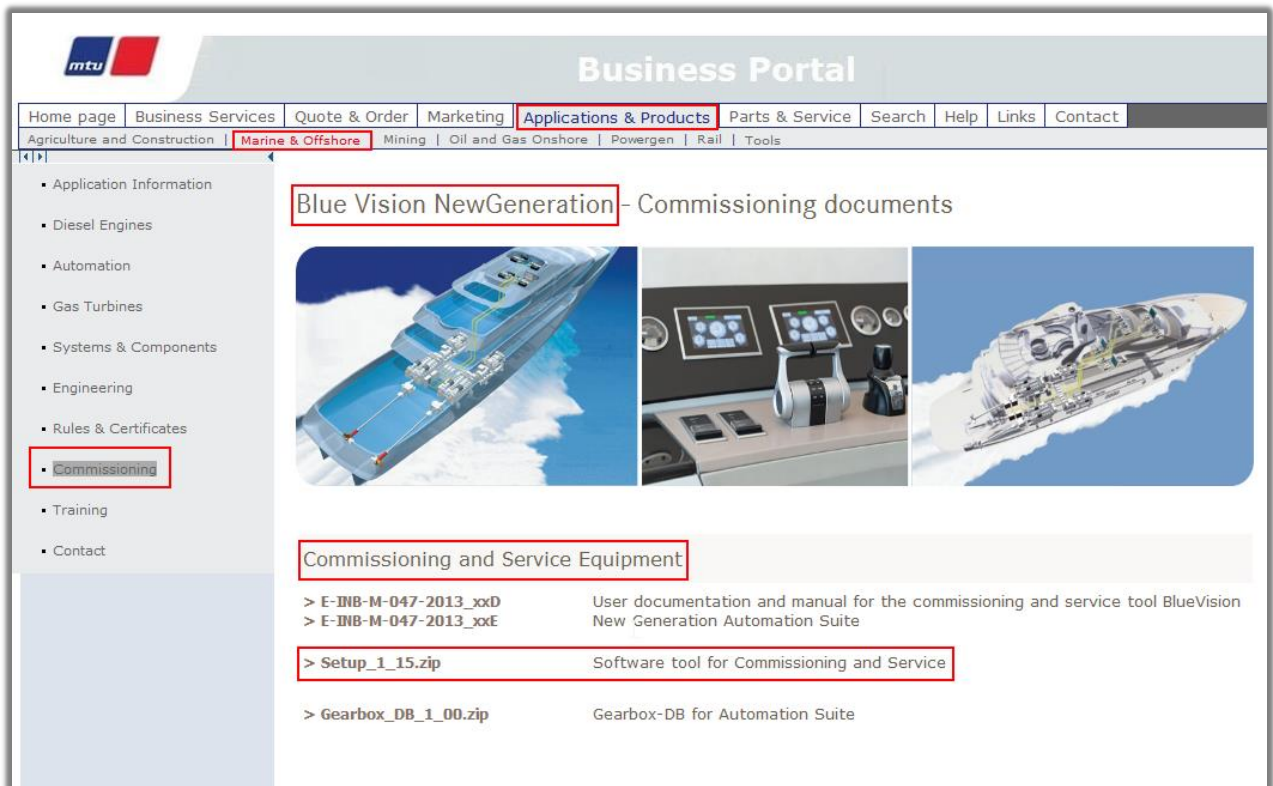


Navigate to the item “**Animate control elements and elements within a screen**” and **deactivate this option** by left-click and confirm by the button “OK”.

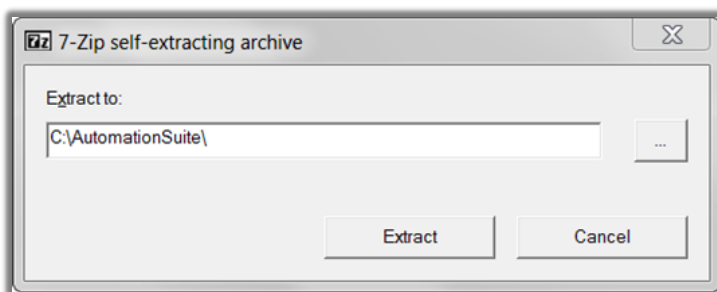
5 Installation

Procedure for the installation of the Automation Suite:

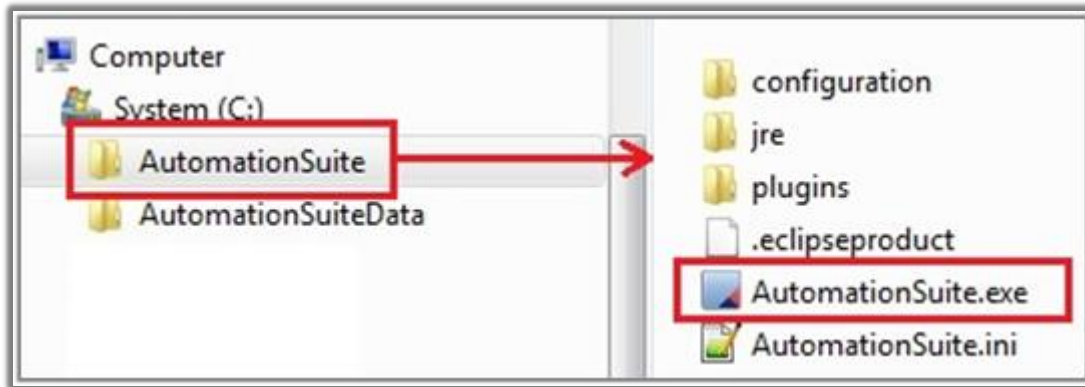
1. Download the setup file from the business portal



2. Execute the setup file and extract to the installation directory “C:\AutomationSuite\”



The installation is now completed.

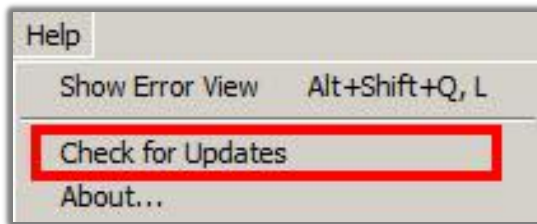


The Automation Suite can be started by the file “AutomationSuite.exe”.

Note: A shortcut of the file “AutomationSuite.exe” (e.g. on the desktop of the service laptop) must be created manually.

6 Update

Submenu item “**Check for Updates**” in menu “**Help**”.



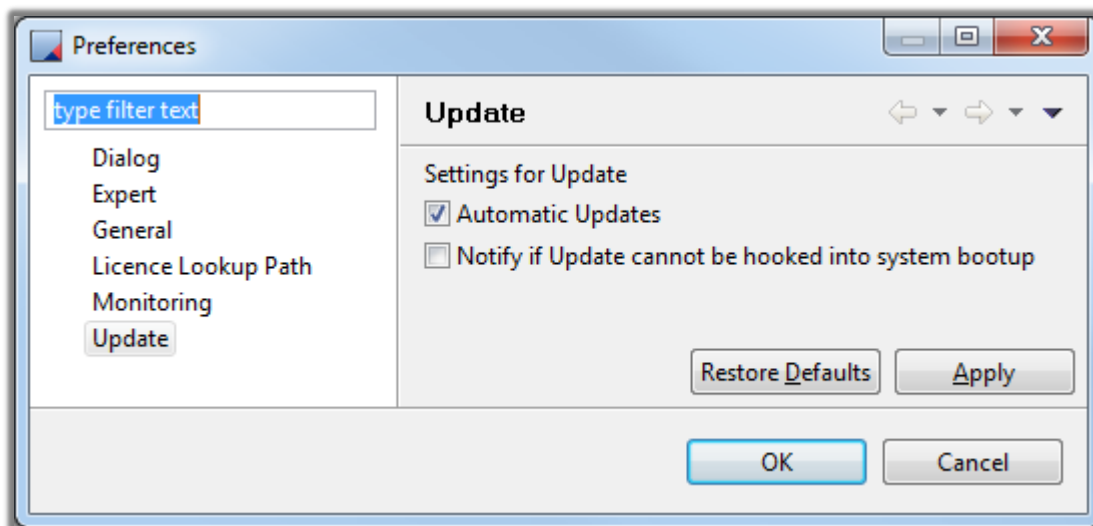
This function can be used to check whether there is a new version of the Automation Suite available. Internet access is required to complete the update process. A dialog will appear if a new version is found. Use the button “Yes” to download and install the new Version. This process may take some time, depending on the speed of the Internet connection. During the update the Automation Suite cannot be used and will be closed.

Note: A backup of the last installed version is saved in the subdirectory “LastVersion”.

Automatic update:

In “Update” under “**Extras/Preferences**” there is an option to disable automatic update search when the Automation Suite is started. If automatic updates are enabled, updates will also be searched at startup of the service laptop.

An alert indicating that the batch files necessary for this cannot be created can be deactivated in the settings.



7 Licence

The user concept for the Automation Suite requires a licence in the form of a file. A licence is user related, has permission attributes and a fixed term.

Note: The system time of the service laptop may not be changed as otherwise the licence might be disabled.

Licence files must be saved on the service laptop in the directory “C:\AutomationSuiteData\Licence”. See chapter 19 *Directory Structure*.

7.1 Activate Licence

If the licence is not activated a dialog is shown after the start of the Automation Suite for entering a password. The password must be defined by the user and should protect against unauthorized access to the licence file.

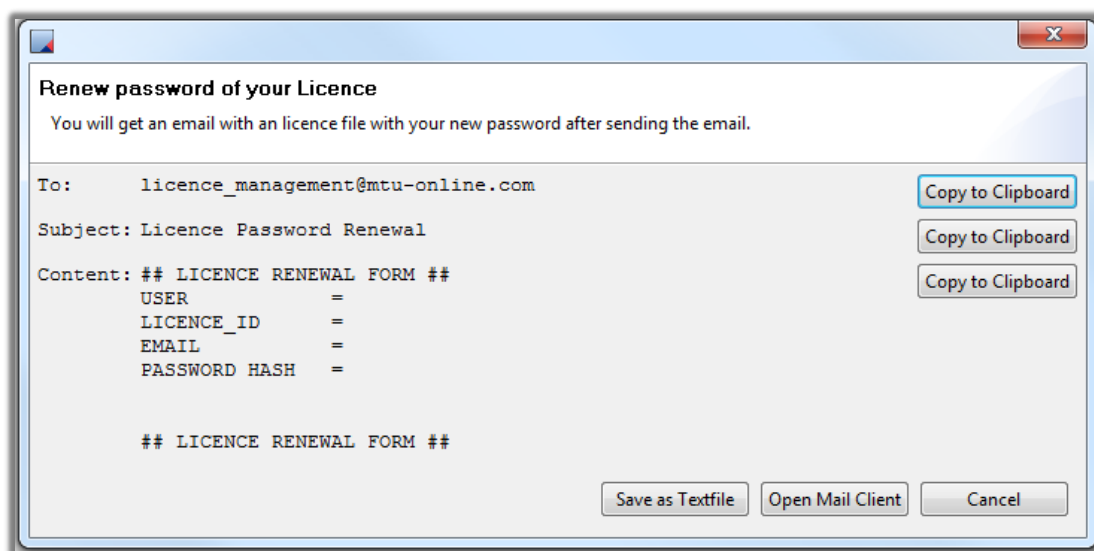


A screenshot of a Windows-style dialog box titled "Welcome". The dialog has a light gray background and a thin border. Inside, the text "Please enter your password" is displayed above two empty text input fields. At the bottom of the dialog, there are three buttons: "Activate", "Work as Guest", and "Select License".

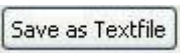
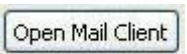
Password conventions:

- Minimum length: 6 characters
- At least one character each from the categories letter, number and special characters

When the button “**Activate**” has been clicked, the password will be verified regarding the required conventions and consistency of the two entries. If the verification has been successful, a new dialog appears which contains the “activation data”. This “activation data” must be sent by email to licence_management@mtu-online.com.



The following buttons are available:

Button	Explanation
	The appropriate text is copied to the clipboard.
	All “activation data” is copied in a text file which can be saved to create and send an e-mail at a later date.
	If an e-mail program is installed, an e-mail with the “activation data” is automatically created which can be saved as a draft or directly sent in case there is an existing internet connection.

After the e-mail has been successfully sent, the user will receive a new licence file via e-mail within a few minutes. The new licence file must be saved in the directory “**C:\AutomationSuiteData\Licence**”.

Note: The original licence file is not required anymore and can be overwritten.

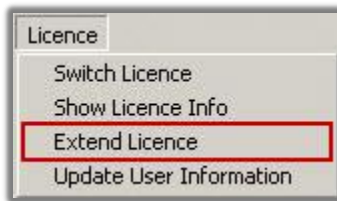
Any change in the licence file directory is shown after the Automation Suite is restarted.

Note: A licence file can be renamed if required.

7.2 Extend Licence

Each licence is only valid for a certain period of time. Thirty days prior to the expiration of the validity the user is notified how many days remain, whenever the licence file is used.

In order to extend the validity of the licence file select the submenu item “**Extend Licence**” in menu “**Licence**”.

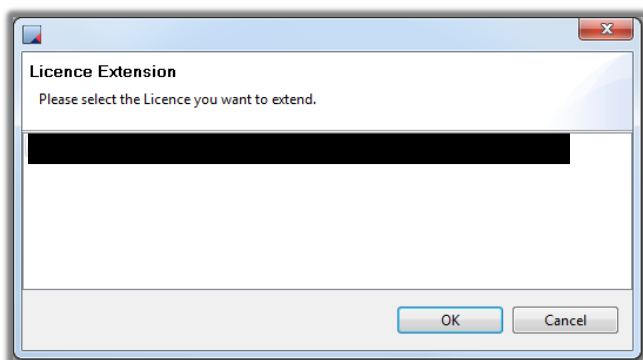


All available licence files are shown in the next dialog.

The background colors indicate the status of the licence file.

Background color	Explanation
	The licence file is valid and not yet activated
	The licence file is valid and already activated
	The licence file is valid but cannot be extended
	The licence file is locked

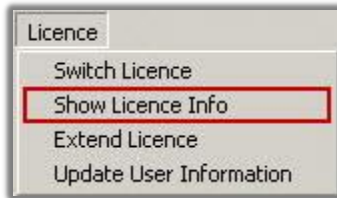
Select the licence file to extend and confirm by the button “OK”.



Continue as described in the licence activation chapter (see chapter *7.1 Activate Licence*).

7.3 View Licence Information

Submenu item “**Show Licence Info**” in menu “**Licence**” or the **toolbar** icon “**Show Licence Information**” can be used to view information of the currently used licence file.

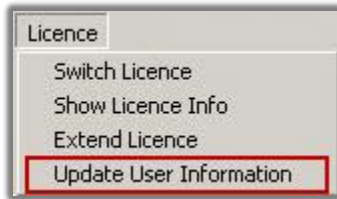


The following Information is displayed:

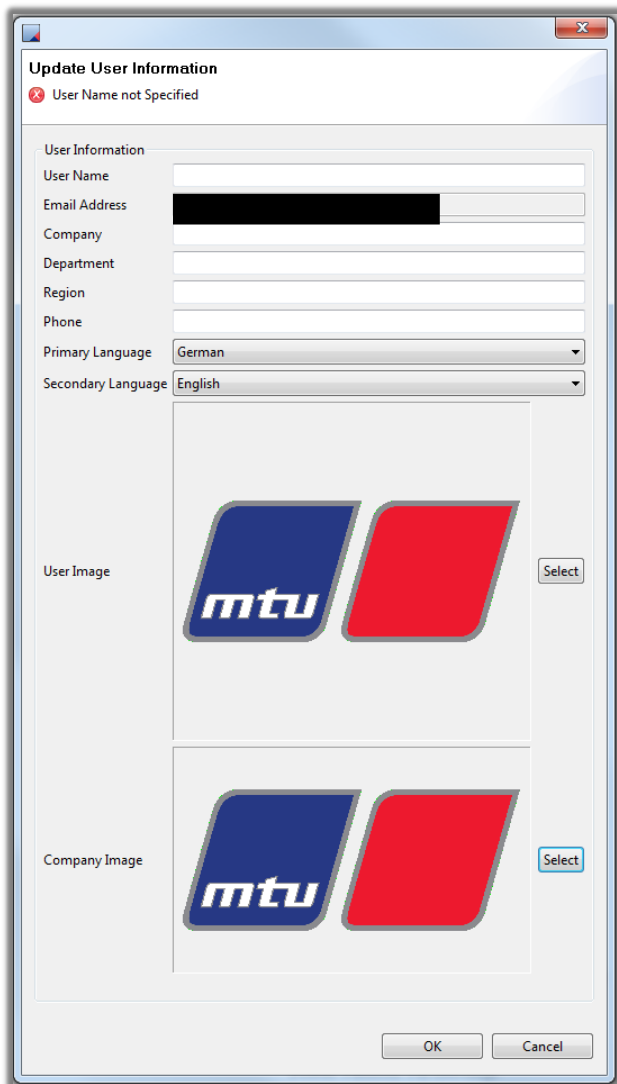
- Personal data
- Validity period
- Supported product lines

7.4 Update Licence Information

The user data of a licence can be changed via the submenu item “**Update User Information**” in menu “**Licence**”.



All user information except the e-mail address can be changed.

A screenshot of the 'Update User Information' dialog box. The dialog has a title bar with a close button. Below the title bar, there is a red error icon and the text 'User Name not Specified'. The main area contains a 'User Information' section with the following fields: 'User Name' (empty), 'Email Address' (redacted with a black box), 'Company' (empty), 'Department' (empty), 'Region' (empty), 'Phone' (empty), 'Primary Language' (dropdown menu set to 'German'), and 'Secondary Language' (dropdown menu set to 'English'). Below these fields are two image selection sections: 'User Image' and 'Company Image'. Each section displays the MTU logo and a 'Select' button. At the bottom of the dialog are 'OK' and 'Cancel' buttons.

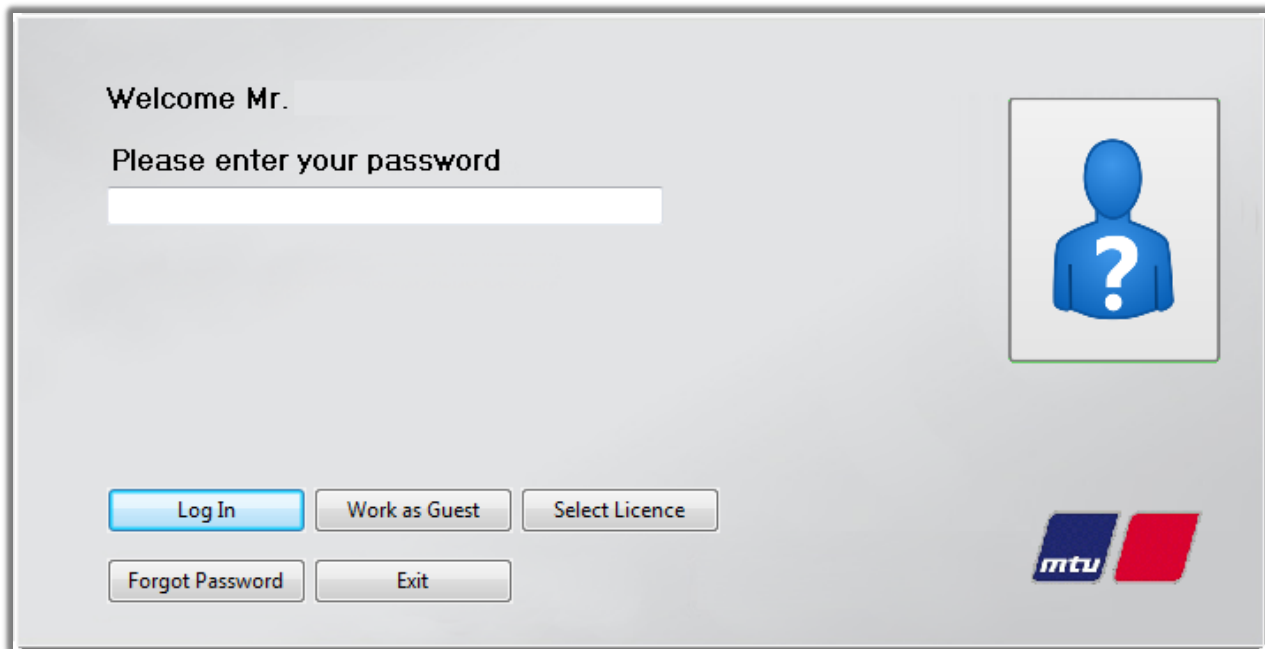
Note: Pictures are supported in the file format PNG, JPG and BMP.

Confirm by the button “OK” and continue as described in the licence activation chapter (see chapter 7.1 *Activate Licence*).

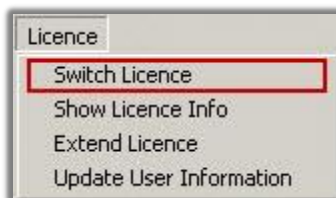
7.5 Switch Licence

In the login dialog the button “**Select Licence**” allows to switch to a different licence. The available licence files are listed in the next dialog where the selection can be done.

Note: At the start of the Automation Suite the last used licence file will be automatically selected if the licence file is still valid.

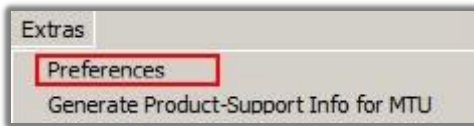


If already logged in the submenu item “**Switch Licence**” in menu “**Licence**” can be used for switching to the login dialog.

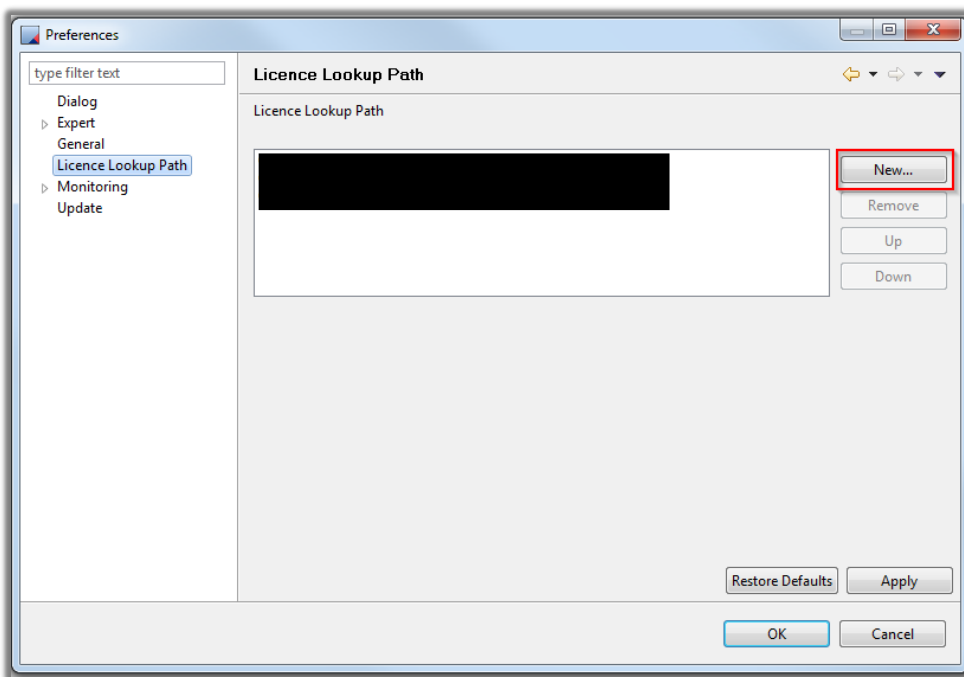


7.6 Licence Directory

The setting “**Licence Lookup Path**” is used to add a licence directory. This setting is available in the submenu item “**Preferences**” in menu “**Extras**”.



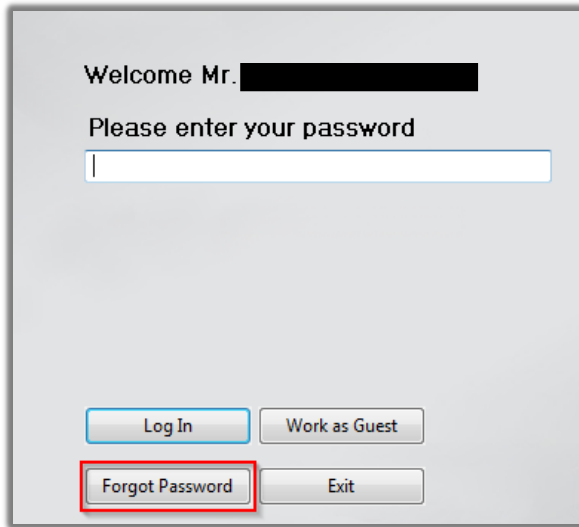
A new directory path can be added via the button “**New**”.
The settings are saved after the button “**OK**” was clicked.



Note: The new settings are activated when the Automation Suite is restarted.

7.7 New Licence Password

With the button “**Forgot Password**” in the login dialog it is possible to define a new password.

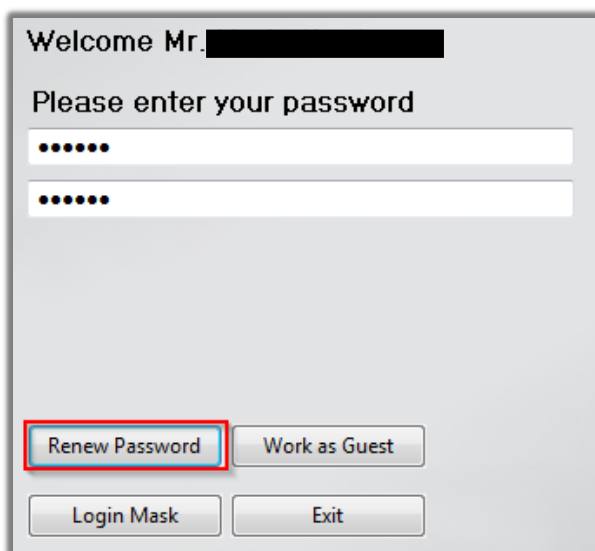


Welcome Mr. [REDACTED]

Please enter your password

The new password must be entered and confirmed by the button “**Renew Password**”.

Continue as described in the licence activation chapter (see chapter 7.1 *Activate Licence*).

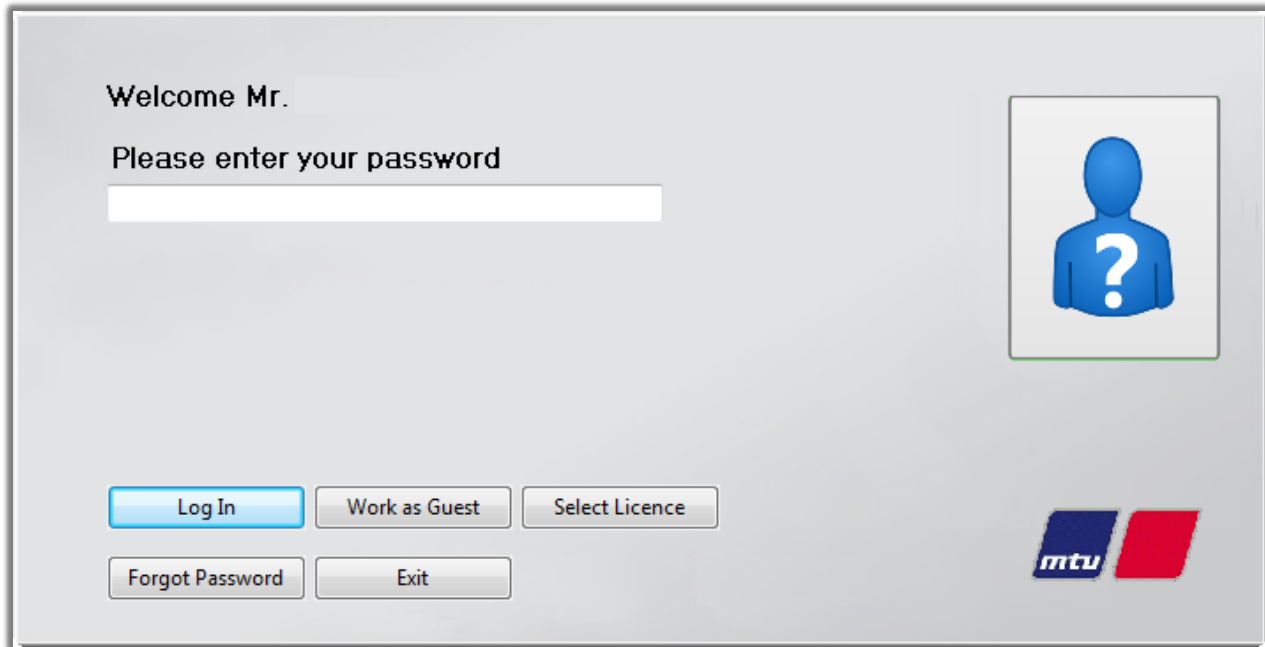


Welcome Mr. [REDACTED]

Please enter your password

8 Login

The Automation Suite is searching for a valid licence file at every start. The valid licence file will be automatically selected and the login dialog appears. In this dialog the password for the licence file can be entered.



In the following cases the login will be done automatically as “guest”(minimum permissions):

- Login dialog is canceled (e.g. with the key “ESC” on the keyboard)
- Button “**Work as Guest**” is selected
- No valid licence file found

9 Connection

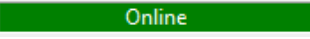
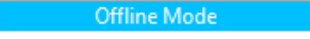
9.1 Online

There must be a connection to at least one SPU node for unlocking functions in the Automation Suite which require an online connection.

A connection can be established using the **toolbar** icon “**Connect to Systembus**”.



The **connection status** is shown in the **status bar**.

Status	Explanation
	No connection has been established to the systembus or the connection is disconnected
	A connection has been established to the systembus and none of the projected SPU nodes are missing.
	Offline-Mode is active (see chapter 9.2 <i>Offline-Mode</i>)
	A connection has been established to the systembus but one or more projected SPU nodes are missing. The missing SPU nodes are displayed in the tooltip.

9.2 Offline-Mode

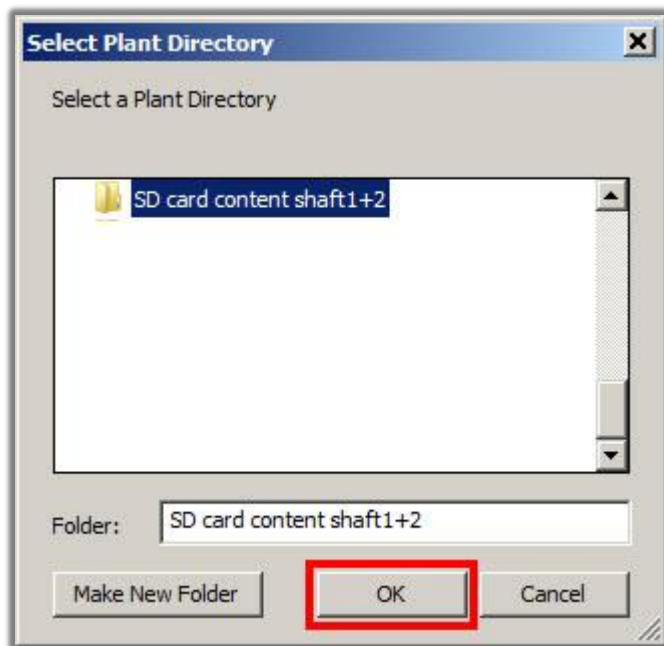
For the “Offline-Mode” various software files are required and must be available locally.

The software files can be copied from the SD-card(s) to the service laptop or via import function from the SPU device directly (see chapter 16.1 *Import Data for Offline-Mode*).

Select the **toolbar** icon “**Load BVNG**” for starting the “**Offline-Mode**”.



In the next dialog the directory with the required software files can be selected.

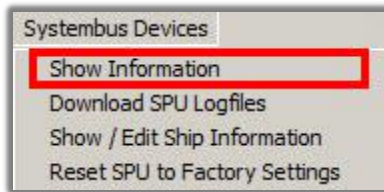


Once the button “**OK**” is clicked the software files will be loaded.

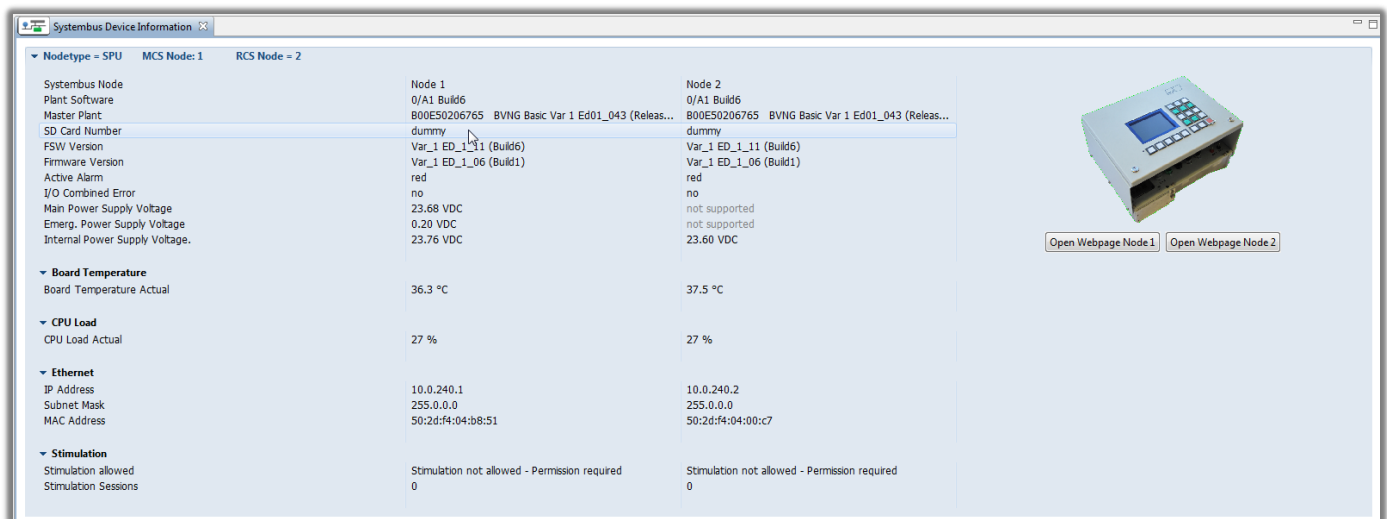
After successful loading the **Offline-Mode** is active.

10 Information about Systembus Devices

Submenu item “**Show Information**” in menu “**Systembus Devices**” or toolbar icon “**Show Device Status**”.
Not available Offline.



The function provides an overview page with all active systembus devices (e.g. SPU, SCU).



The button “**Open Webpage**” opens the website of this SPU node directly in the Automation Suite.

The button “**More**” show additional information specific to the device.

11 Logfiles

11.1 Download SPU Logfiles

Submenu item “**Download SPU Logfiles**” in menu “**Systembus Devices**”.
Not available Offline.



The submenu item “**Download SPU Logfiles**” saves the logfiles of the connected SPU nodes in a local directory (defined by the user). The logfiles are ordered according to shafts and their SPU nodes.
Once the download has been completed the directory is opened with the Windows Explorer automatically.



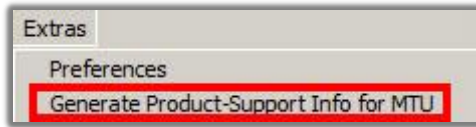
Each SPU node provides the following **logfiles**:

File name	Explanation
alarm.log.csv	Alarm History
PARAMS.LOG.CSV	History of all parameter changes by the user
PARAMS_SPS.LOG.CSV	History of all parameter changes by the SPC

Note: The logfiles are also included in the data retrieval file (see chapter 16.1 *Create Data Retrieval File*).

11.2 Generate Product Support Info for MTU

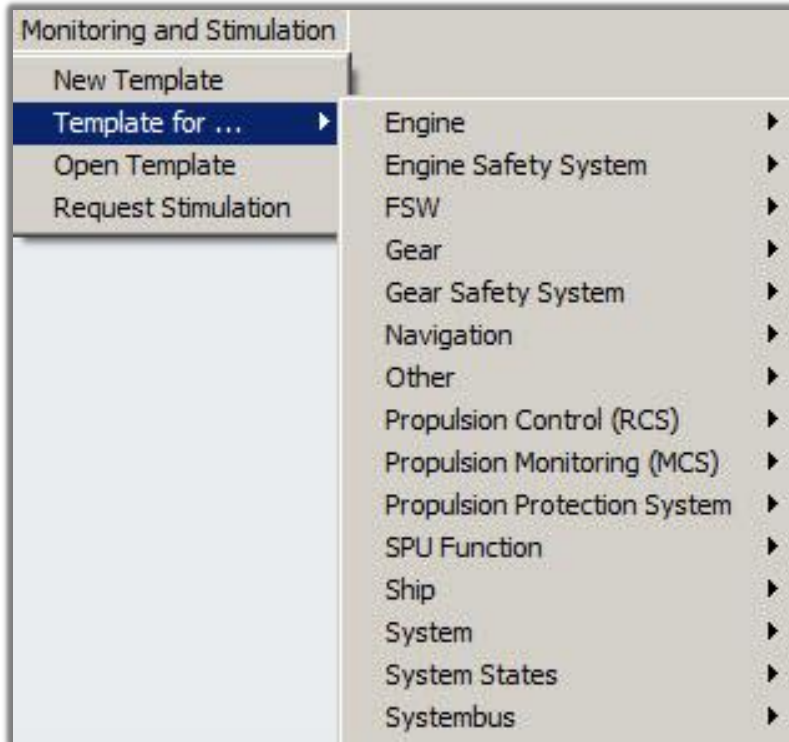
Submenu item **“Generate Product Support Info for MTU”** in menu **“Extras”**.
Not available Offline.



This function saves data about the current system status into an encrypted file which can be forwarded to MTU in case of an issue.

12 Monitoring and Stimulation

Menu “Monitoring and Stimulation”



Important: Changes may only be made by authorized personnel. Proceeding incorrectly may result in damage to persons or the machine. Therefore a warning is displayed at the beginning which has to be confirmed by the user.

Note: The warning is displayed once per session

12.1 Tabular View

Shaft 1 (26)										
ID	Node	Name	Value	Unit	MD	SD	Plot	Record	Stimulate	
1.001.0010.04330	Node 1	Cmd_ResetEmergencyStop	☐	Bool	☐	☐	☐	☐	☐	
1.001.0010.04330	Node 1	AmountofMonitEmergStopPB	1	digit	☐	☐	☐	☐	☐	
1.001.0010.04330	Node 1	StatusEmergStopCircuitFail	MD	digit	☒	☐	☐	☐	☐	
1.001.0010.04330	Node 1	ST_AL_EmergStopCircuitFail	☒	Bool	☒	☐	☐	☐	☐	
1.001.0010.04330	Node 1	AL_EmergStopCircuitFail	0	digit	☐	☐	☐	☐	☐	
1.001.0010.04330	Node 1	ST_WB_ESIECU	☒	Bool	☒	☐	☐	☐	☐	
1.001.0010.04330	Node 1	WB_ESIECU	0	digit	☐	☐	☐	☐	☐	
1.001.0010.04330	Node 1	ST_AL_EmergencyStop	☒	Bool	☒	☐	☐	☐	☐	

In its default setting, the tabular view of “**Monitoring and Stimulation**” has following columns:

- Plot (see chapter 12.5 *Plot screen*)
- Record (see chapter 14.1 *Tabular View*)
- Stimulate (see chapter 12.6 *Stimulation*)

12.2 Predefined Templates

Submenu item “**Template for...**” in menu “**Monitoring and Stimulation**”.

The entries within the submenu item “**Template for...**” correspond to **System groups** (e.g. Gear Safety System). System groups are subdivided into other **Subcategories** (e.g. P-Control Oil). Selecting such an item opens a tabular view with all SysIDs of this subcategory.

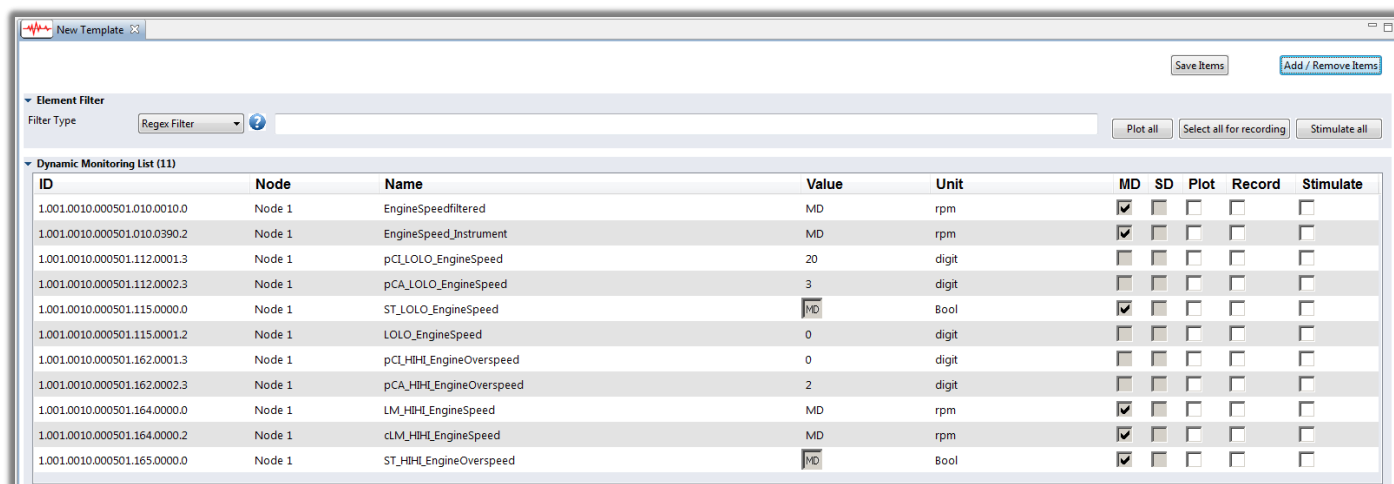
Note: The entry “Commissioning” is not related to a system group.

12.3 New Template

Submenu item “**New Template**” in menu “**Monitoring and Stimulation**” or toolbar icon “**New Stimulation Template**” opens a tabular view without SysIDs (empty).



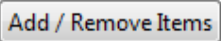
The function “**New Template**” allows to create an individual template.



The screenshot shows the 'New Template' dialog box with a 'Dynamic Monitoring List (11)' table. The table has columns for ID, Node, Name, Value, Unit, MD, SD, Plot, Record, and Stimulate. The 'MD' column has checkboxes, and the 'Plot', 'Record', and 'Stimulate' columns have buttons.

ID	Node	Name	Value	Unit	MD	SD	Plot	Record	Stimulate
1.001.0010.000501.010.0010.0	Node 1	EngineSpeedfiltered	MD	rpm	☒	☐	☐	☐	☐
1.001.0010.000501.010.0390.2	Node 1	EngineSpeed_Instrument	MD	rpm	☒	☐	☐	☐	☐
1.001.0010.000501.112.0001.3	Node 1	pCL_LOLO_EngineSpeed	20	digit	☐	☐	☐	☐	☐
1.001.0010.000501.112.0002.3	Node 1	pCA_LOLO_EngineSpeed	3	digit	☐	☐	☐	☐	☐
1.001.0010.000501.115.0000.0	Node 1	ST_LOLO_EngineSpeed	MD	Bool	☒	☐	☐	☐	☐
1.001.0010.000501.115.0001.2	Node 1	LOLO_EngineSpeed	0	digit	☐	☐	☐	☐	☐
1.001.0010.000501.162.0001.3	Node 1	pCL_HiHi_EngineOverspeed	0	digit	☐	☐	☐	☐	☐
1.001.0010.000501.162.0002.3	Node 1	pCA_HiHi_EngineOverspeed	2	digit	☐	☐	☐	☐	☐
1.001.0010.000501.164.0000.0	Node 1	LM_HiHi_EngineSpeed	MD	rpm	☒	☐	☐	☐	☐
1.001.0010.000501.164.0000.2	Node 1	clM_HiHi_EngineSpeed	MD	rpm	☒	☐	☐	☐	☐
1.001.0010.000501.165.0000.0	Node 1	ST_HiHi_EngineOverspeed	MD	Bool	☒	☐	☐	☐	☐

The following buttons are available:

Button	Explanation
	Opens the dialog to add and remove SysIDs.
	Saves the current template in a file.

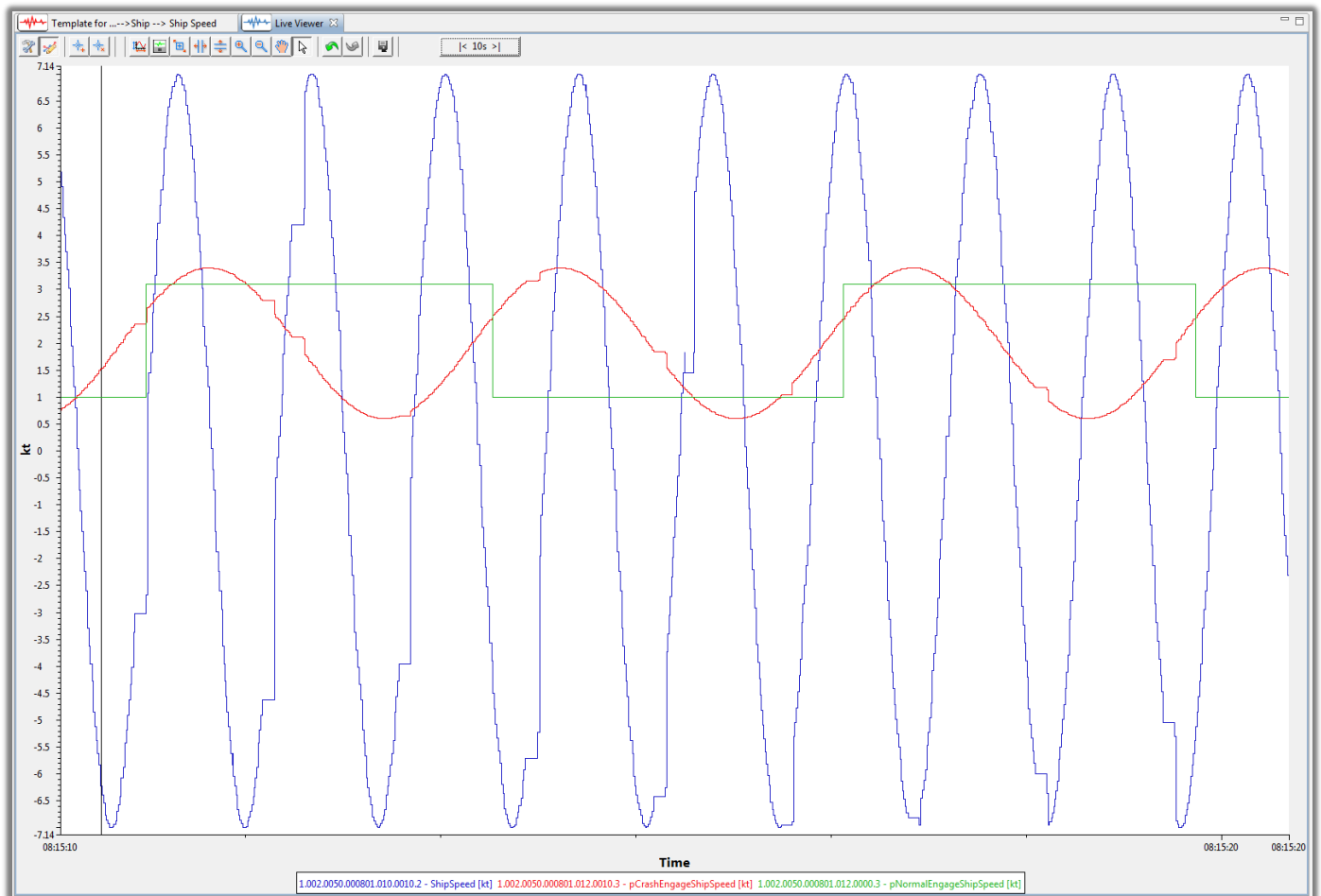
12.4 Open Template

Submenu item “**Open Template**” in menu “**Monitoring and Stimulation**” or toolbar icon “**Open Stimulation Template**” allows to open an individual template.



12.5 Plot screen

The Plot screen is used to graphically output recorded and monitored SysIDs.
For this, the selected SysIDs are depicted in a **line diagram** which shows the values on a shared time axis.



12.5.1 Functions



The functions are:

1. **Options** (see chapter 12.5.2 Options)
2. **Fading legend in/out**
3. **Add annotation** (see chapter 12.5.3 Markers (Annotations))
4. **Remove annotations**
5. **Auto zoom**

The Auto zoom changes the view to fit the available curves into the window completely

6. Start/Pause

Starts or pauses output of the Plot screen

7. Rectangular zoom (mouse selection)

Zoom into rectangular area

8. Horizontal zoom (mouse selection)

Zoom into horizontal area

9. Vertical zoom (mouse selection)

Zoom into vertical area

10. Zoom In

Allows zooming into a selected point

11. Zoom Out

Allows zooming out from a point

12. Move

Moving of the view point

13. Normal cursor (without function)

14. Undo (change view)

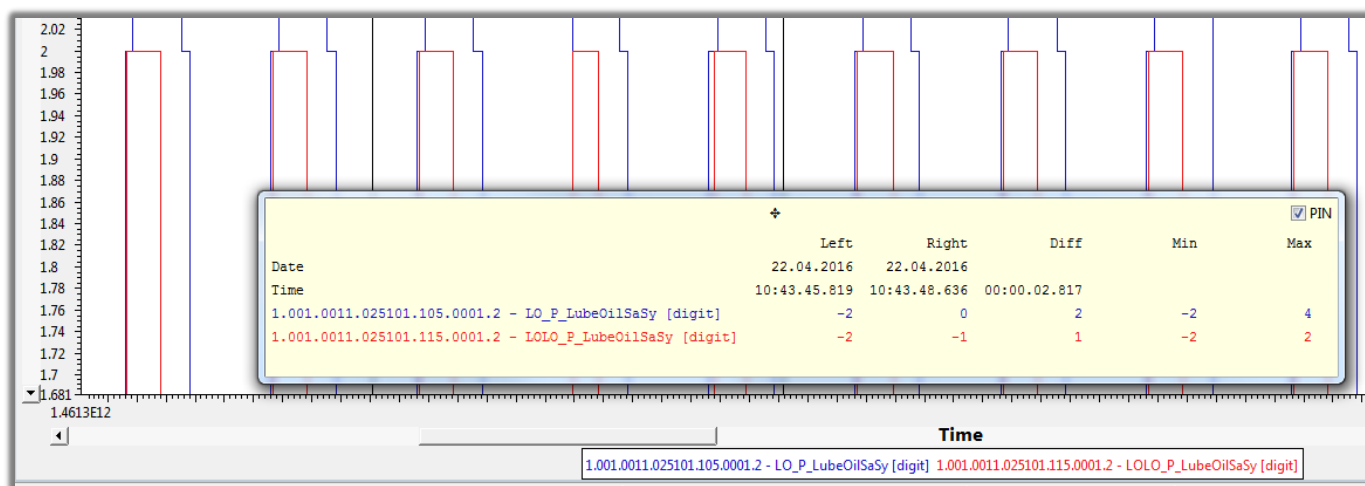
15. Restore (change view)

16. Print/save window screen

Likewise, if CSV or h5 format is selected as the file type, only the current window screen is saved

17. Set view timespan (only in case of Live Display)

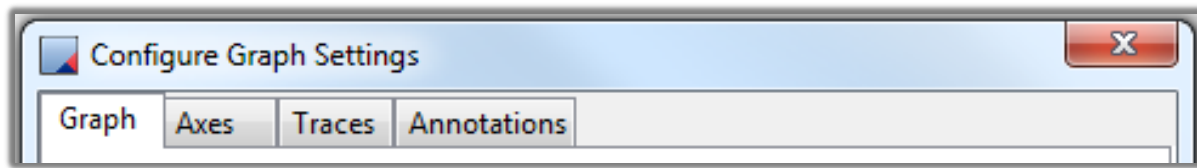
Two cursors (measurement lines) are available for evaluation. The cursors can be moved to the desired position by holding down the left mouse button and moving the mouse. Moving the mouse to one of the two cursors displays a tool tip which represents the values of both cursors and their difference.



Note: A SysID can be temporarily shown/ hidden, by double clicking the SysID entry below the graphic screen.

12.5.2 Options

The option dialog for the Plot screen consists of four tabs, as described below.



Note: “Annotations” tab only becomes visible after an annotation has been added, e.g. in Plot with  .

Graph:

Fonts, font colors and a background color can be set under “Graph”.
It is also possible to show or hide parts of a screen.

Axes:

Various settings for the available axes can be selected.
These include:

- Change title name
- Change axis color
- Activate/deactivate autoscaling
- Activate/deactivate grid lines

Traces:

Settings for the displayed curves of the SysIDs can be selected under “Traces”.
Setting options include the following:

- Adapt name
- Unit selection for X and Y axes
- Curve width change
- Curve color change
- Hide/Show

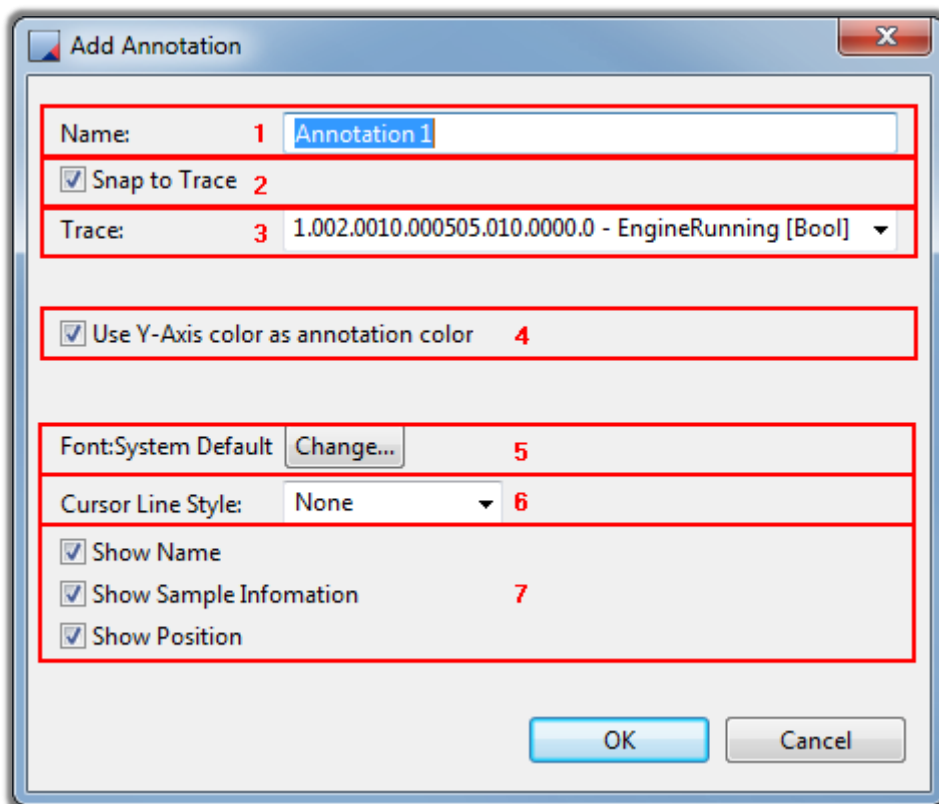
Annotations:

(See *chapter 12.5.3 Markers (Annotations)*):

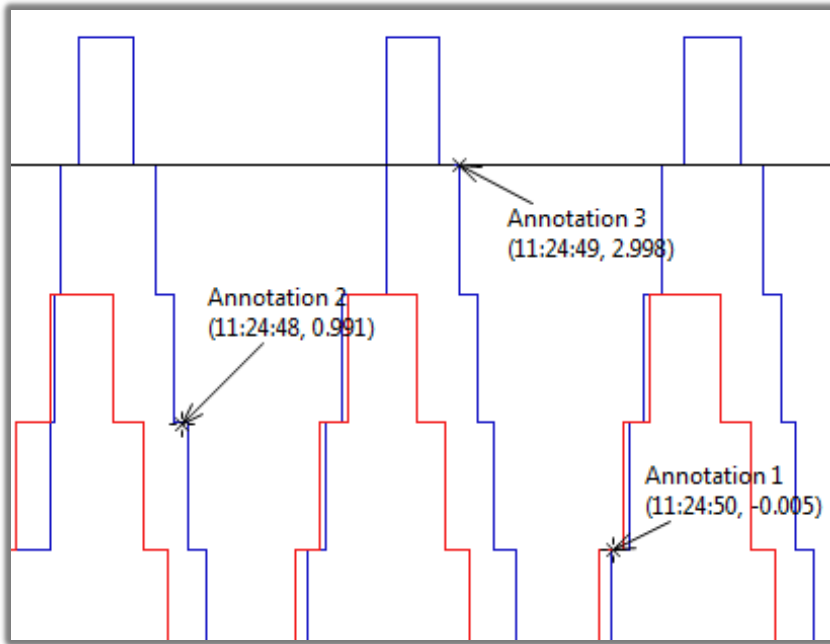
12.5.3 Markers (Annotations)

It is possible to add markers for curves within the Plot screen.

Clicking on the **“add Annotation”** (see chapter 12.5.1 *Functions*) opens a dialog, described below.



1. The annotation name
2. The **“Snap to Trace”** selection defines whether the annotation is to be pinned to a signal or floating. If “Snap to Trace” is active, a SysID to which the annotation is connected must be selected. If deactivated, selection of a coordinate system based on the X and Y axis labeling must be selected. In this setting, the annotation is floating.
3. SysID on which the marker shall be snapped.
4. Selection whether Y axis color is also be used for annotations. If this is deactivated, custom color can be selected.
5. Font definition
6. The appearance of the annotation can be defined more precisely
7. Definition of further annotation properties



Curve with annotations

Active annotations can be edited by double clicking on its position.

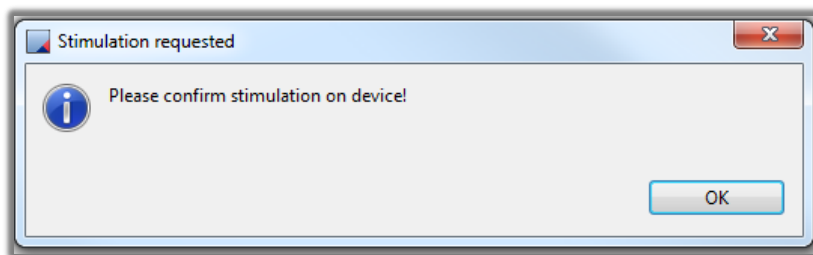
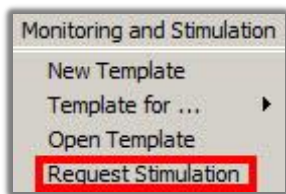
Note: Annotations are also saved when saving as H5 file.

12.6 Stimulation

12.6.1 Stimulation Permission

Before a SysID can be stimulated the “stimulation permission” must have been confirmed on the SPU service-display. Before the “stimulation permission” can be confirmed on the SPU service-display it is necessary to activate a SysID of this SPU in the “**Stimulate**” column.

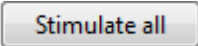
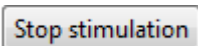
Alternatively, the “stimulation permission” can be requested via the submenu item “**Request Stimulation**” in the menu “**Monitoring and Stimulation**”. In this case a “Stimulation permission” request is send to all connected SPU(s).



To confirm the “stimulation permission” for a SPU the user must enter the “Stimulation” page in the service-display and confirm the entry “**Allow Stimulation Request**” by pressing the “**Enter**” key. After the stimulation request has been confirmed all SysIDs of this SPU can be stimulated.

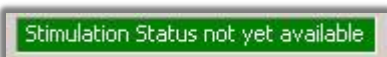
By selecting a SysID in the column “**Stimulate**” the stimulation of this SysID starts with the last active value. This value can be overwritten in the column “Value”.

As soon as the stimulated SysID is deselected in the column “Stimulate” the stimulation of this SysID is stopped.

Pressing button  or  allows all SysIDs of this tabular view to be stimulated at once or stimulation to be stopped respectively.

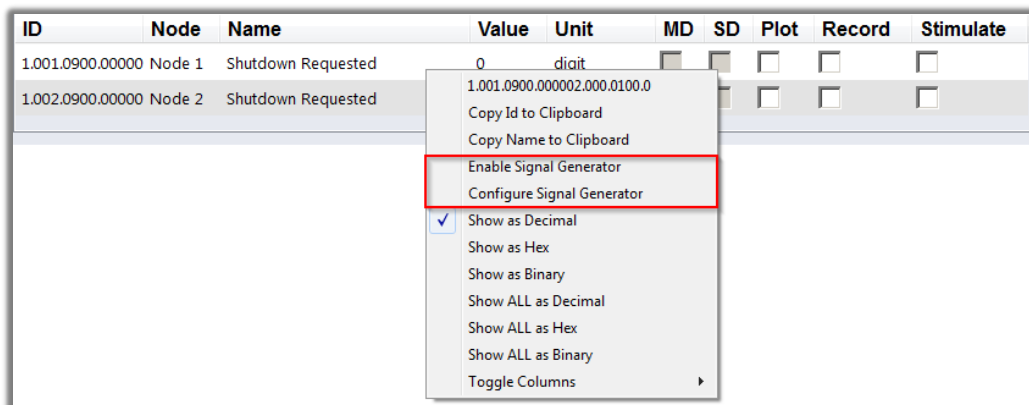
Note: The stimulation permission for a SPU is revoked if no SysID of this SPU is stimulated for 5 minutes. If the tabular view is closed the stimulation of these SysIDs is stopped.

The **stimulation status** is available in the **status bar**:

Status	Explanation
	Disclaimer not confirmed
	Stimulation on all SPUs allowed
	Stimulation not confirmed on all SPUs
	Stimulation not confirmed on any SPU
	Stimulation status not available
	One or more SysIDs are stimulated

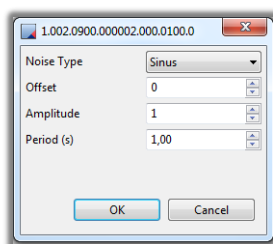
12.6.2 Signal generators

To access the function “Signal Generator” the context menu must be opened (see chapter 18.1.1 *Context menu in the tabular view*). Before the signal generator can be started via the entry “**Enable Signal Generator**” the entry “**Configure Signal Generator**” must be used to configure the signal generator.



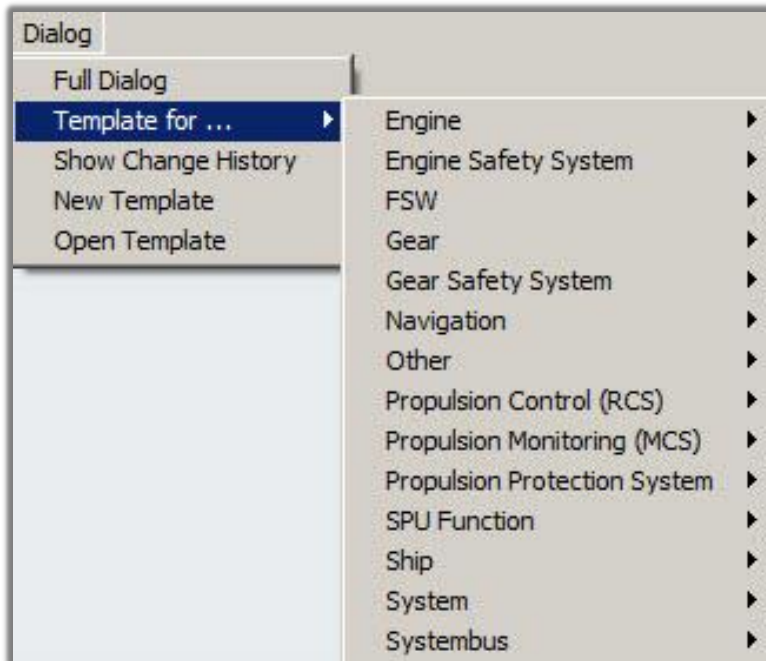
The following **signal types** can be configured:

- Sinus
- Ramp
- Triangle
- Rectangle
- Random



13 Dialog

Menu “**Dialog**”.



The Dialog function allows the permanent modification of SysID parameters within the system.

Important: Changes may only be made by authorized personnel. Proceeding incorrectly may result in damage to persons or the machine. Therefore a warning is displayed at the beginning which has to be confirmed by the user.

Note: The warning is displayed once per session.

13.1 Full Dialog

Submenu item “**Full Dialog**” in menu “**Dialog**” or **toolbar** icon “**Full Dialog**” open a tabular view with all SysID parameters.



Full Dialog

Save parameter values to disk Load parameter values from disk Undo Changes Save engineering data

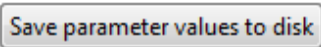
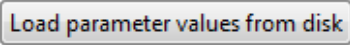
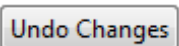
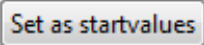
Element Filter

Filter Type: Regex Filter

Shaft 1 (2333)

ID	Node	Name	Value	Unit	MD
1.001.0010.000501.112.0001.3	Node 1	pCL_LOLO_EngineSpeed	2	digit	☐
1.001.0010.000501.112.0002.3	Node 1	pCA_LOLO_EngineSpeed	3	digit	☐
1.001.0010.000501.162.0001.3	Node 1	pCL_HH_HL_EngineOverspeed	0	digit	☐
1.001.0010.000501.162.0002.3	Node 1	pCA_HH_HL_EngineOverspeed	2	digit	☐
1.001.0010.000501.232.0001.3	Node 1	pCL_MG_EngSpeedHiloClutch	0	digit	☐
1.001.0010.000501.232.0002.3	Node 1	pCA_MG_EngSpeedHiloClutch	39	digit	☐
1.001.0010.000506.162.0001.3	Node 1	pCL_HH_HL_OverspeedCamshaft	0	digit	☐
1.001.0010.000506.162.0002.3	Node 1	pCA_HH_HL_OverspeedCamshaft	2	digit	☐
1.001.0010.000506.212.0001.3	Node 1	pCL_SD_EngineSpeedCamshaft	0	digit	☐
1.001.0010.000506.212.0002.3	Node 1	pCA_SD_EngineSpeedCamshaft	131076	digit	☐
1.001.0010.000507.212.0001.3	Node 1	pCL_SD_EngineSpeedCrankshaft	0	digit	☐
1.001.0010.000507.212.0002.3	Node 1	pCA_SD_EngineSpeedCrankshaft	131076	digit	☐
1.001.0010.000508.010.0020.3	Node 1	pEngineSpeedHalfScaleDIS	MD	rpm	☒
1.001.0010.000508.010.0030.3	Node 1	pEngineSpeedMaxScaleDIS	MD	rpm	☒
1.001.0010.000509.302.0001.3	Node 1	pEngineSpeed_Output_FRCS	Show Values	rpm/V	☐
1.001.0010.000510.104.0000.3	Node 1	pLM_LO_SpeedWindow1	500.0	rpm	☐
1.001.0010.000510.108.0000.3	Node 1	pHY_LO_SpeedWindow1	0.0	rpm	☐
1.001.0010.000510.154.0000.3	Node 1	pLM_HL_SpeedWindow1	1000.0	rpm	☐
1.001.0010.000510.158.0000.3	Node 1	pHY_HL_SpeedWindow1	0.0	rpm	☐
1.001.0010.000511.104.0000.3	Node 1	pLM_LO_SpeedWindow2	500.0	rpm	☐
1.001.0010.000511.108.0000.3	Node 1	pHY_LO_SpeedWindow2	0.0	rpm	☐
1.001.0010.000511.154.0000.3	Node 1	pLM_HL_SpeedWindow2	50.0	rpm	☐
1.001.0010.000511.158.0000.3	Node 1	pHY_HL_SpeedWindow2	0.0	rpm	☐
1.001.0010.000512.012.0000.3	Node 1	pIdleSpeedIncreaseMode	200	digit	☐
1.001.0010.000604.202.0001.3	Node 1	pCL_AL_AnalogSpeedDemand	0	digit	☐
1.001.0010.000604.202.0002.3	Node 1	pCA_AL_AnalogSpeedDemand	4	digit	☐
1.001.0010.000607.232.0001.3	Node 1	pCL_MG_NotLocalDecSpd	0	digit	☐
1.001.0010.000607.232.0002.3	Node 1	pCA_MG_NotLocalDecSpd	39	digit	☐
1.001.0010.000608.232.0001.3	Node 1	pCL_MG_NotLocalIncSpd	0	digit	☐
1.001.0010.000608.232.0002.3	Node 1	pCA_MG_NotLocalIncSpd	39	digit	☐
1.001.0010.002013.302.0001.3	Node 1	pEngineLoad_Output_FRCS	Show Values	%/V	☐

The following buttons are available:

Button	Explanation
	Saves all parameter values of the current tabular view in a file
	Restoring of parameter values from a file to the current tabular view. Note: The tabular view must contain the corresponding SysID(s) and the user must have the appropriate permission of the SysID(s). Therefore it is recommended to restore parameter values the “Full Dialog”.
	The changed parameter values of the current tabular view are reset to the original value. Note: This function will not change back values which are already transferred to the device(s).
	Starts the process of transferring the changed start values of the parameters to the device(s).

13.2 Predefined Templates

Submenu item “**Template for...**” in menu “**Dialog**”.

The entries within the submenu item “**Template for...**” correspond to **System groups** (e.g. Gear Safety System). System groups are subdivided into other **Subcategories** (e.g. P-Control Oil). Selecting such an item opens a tabular view with all SysIDs of this subcategory.

Note: The entry “Commissioning” is not related to a system group.

13.3 New Template

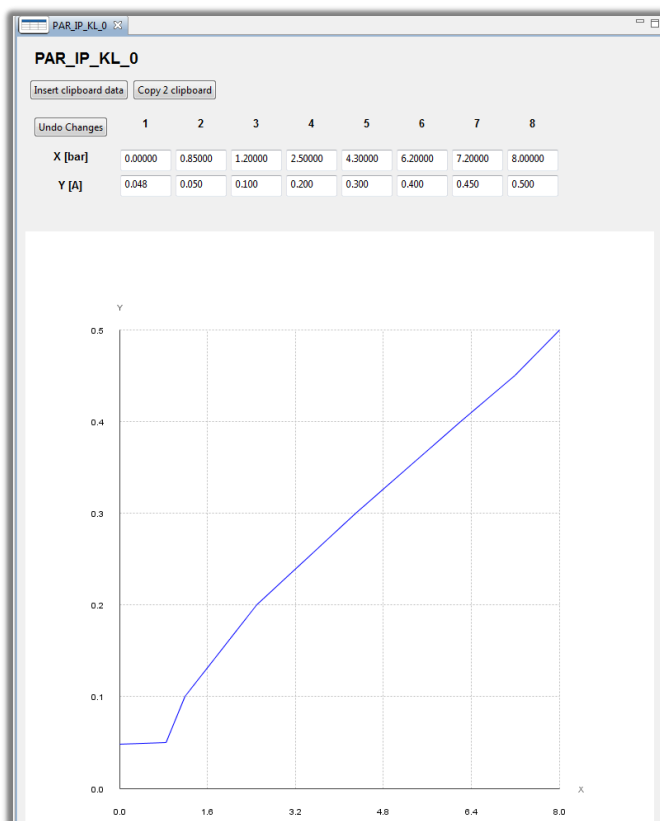
Submenu item “**New Template**” in menu “**Dialog**” open a tabular view without any SysID parameters (empty). This function allows to create an individual template.

13.4 Open Template

Submenu item “**Open Template**” in menu “**Dialog**” open an individual template.

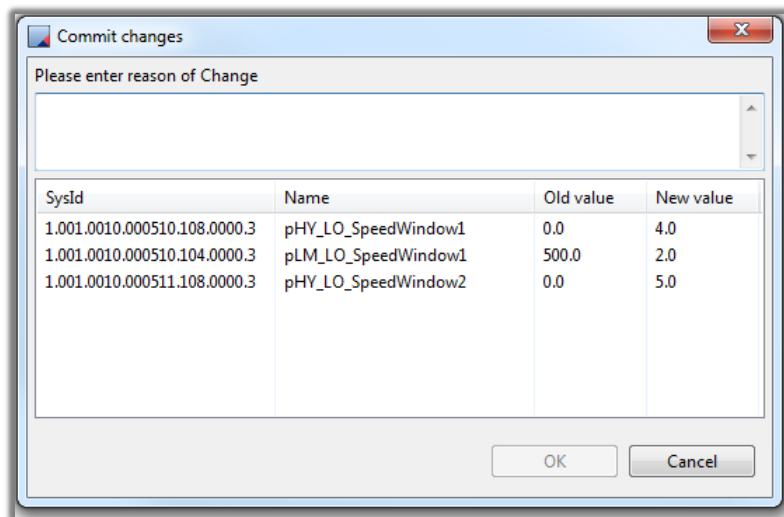
13.5 Multidimensional parameters

For a multidimensional parameter the button “**Show Values**” is displayed in the column “**Start Value**”. The multidimensional parameter is displayed graphically if the button “Show Values” is used.



13.6 Changing Parameters

A parameter value can be changed in the column **“Start Value”** in the tabular view. The changed values can be applied via the button **“Set as start values”**. Before the parameter values are transferred to the device the **“Commit changes”** dialog is displayed in which the parameters to be changed are listed.

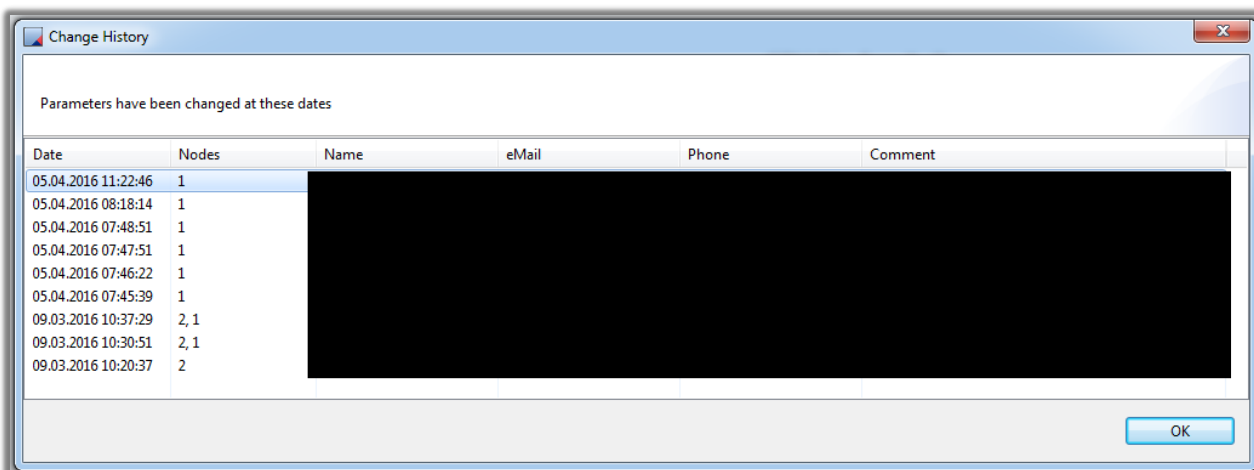


The reason for the parameter change must be entered in the text field **“Please enter reason of Change”** in English. The transfer of the parameters to the device starts after the **“OK”** button has been clicked.

Note: At the beginning of the first parameter transfer (since program start) a recovery point (Parameter Backup) is created automatically. See chapter 16.4 *Recovery Point (Parameter Backup)*

13.7 Parameter Change History

Submenu item **“Show Change History”** is not available in Offline-Mode.



The function **“Show Change History”** shows all parameter transfers since last software installation.

The listed parameter transfers are shown with the following information:

- Date and time
- Affected SPU node(s)
- Licence information
- User comment

Note: The changed parameter values are recorded in the logfiles (see chapter *11 Logfiles*).

14 Recording

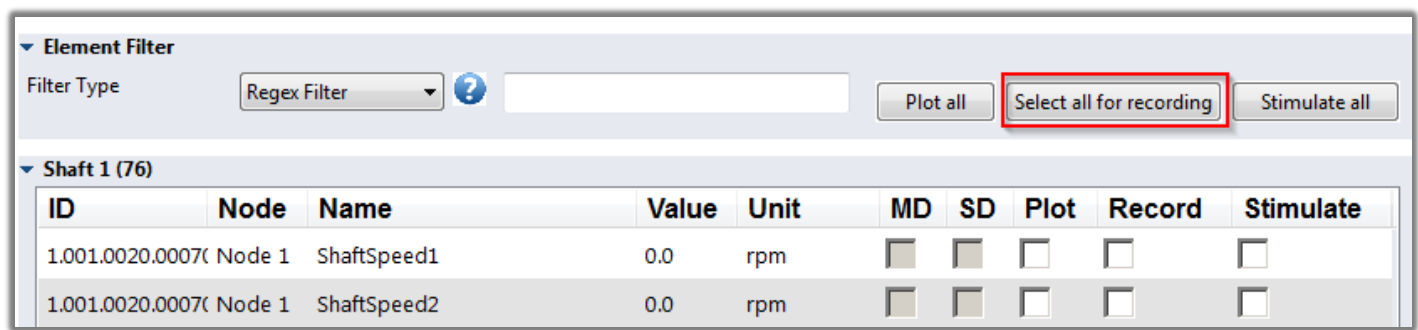
A recording of SysIDs can be done with the submenu item **“Advanced Data Recorder”** in menu **“Data recorder”** or in the tabular view **“Monitoring and Stimulation”**.

Note: The value of a recorded SysID is not visible in the Data Recorder.

14.1 Tabular View

Before the recording in **“Monitoring and Stimulation”** can be started one or more SysID entries must be selected in the column **“Record”**.

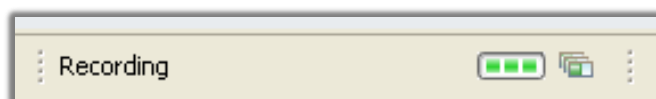
The button **“Select all for recording”** will activate the checkbox in the column **“Record”** for all SysIDs in the tabular view. This button can also be used to deactivate the checkbox for all SysIDs.



Toolbar icon **“Start Recording”** or submenu item **“Start/Stop Recording”** in menu **“Data Recorder”** will start the recording.



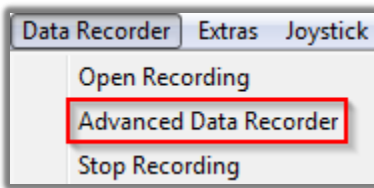
If the button **“Start Recording”** is clicked the folder and file name for the recording file must be defined. Afterwards the recording is active and a status indicates a running recording. The same button or same submenu item is used to stop the recording.



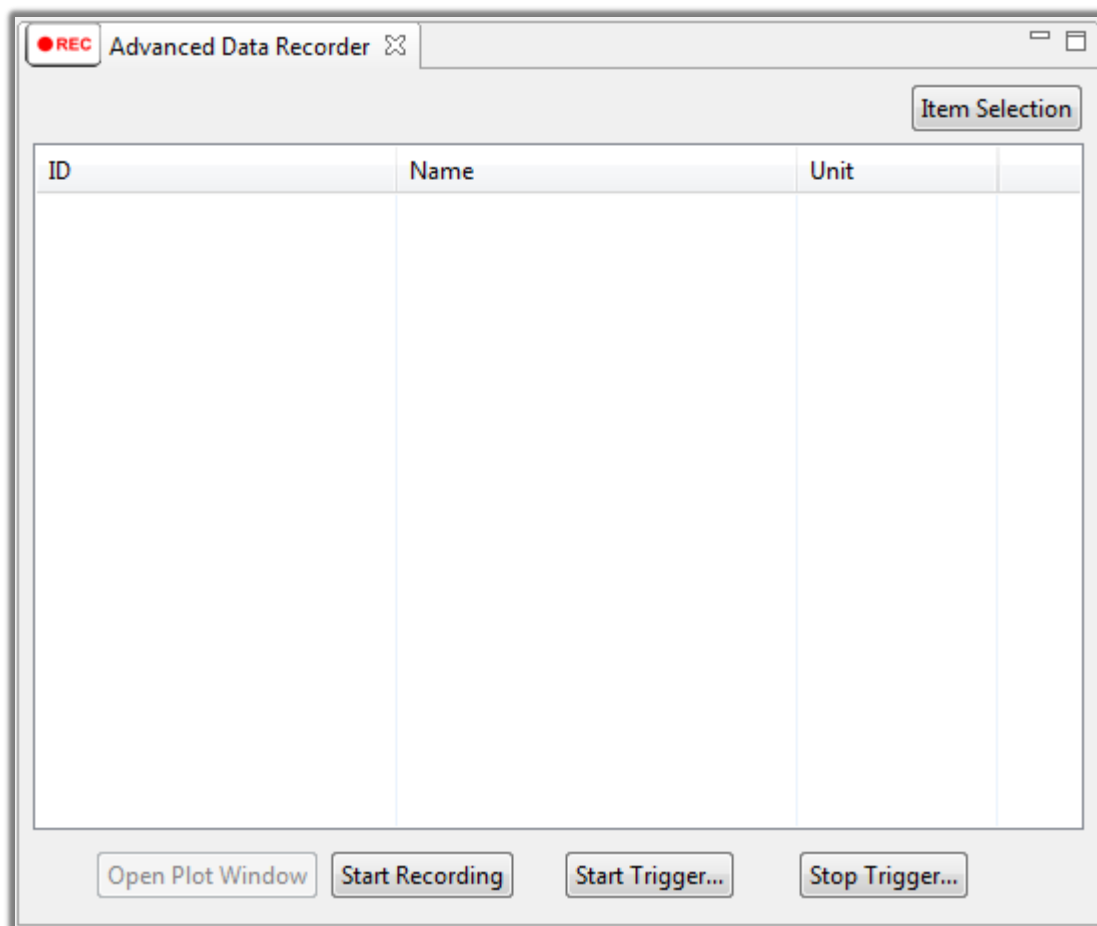
Note: Closing the tabular view will stop the recording.

14.2 Advanced Data Recorder

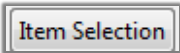
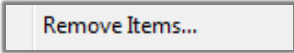
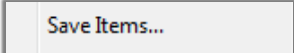
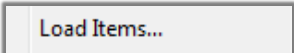
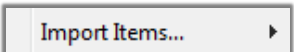
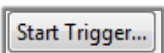
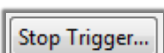
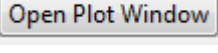
Submenu item “**Advanced Data Recorder**” in menu “**Data Recorder**”.



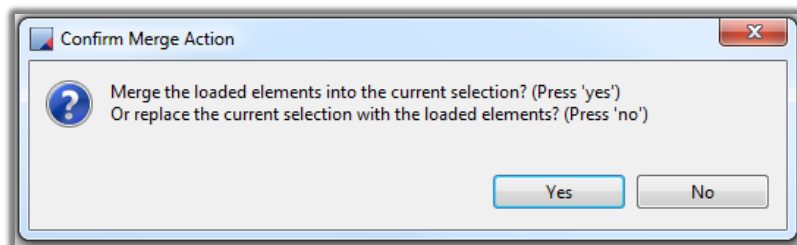
The Advanced Data Recorder will open.



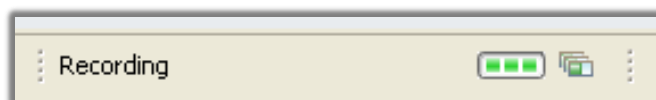
The most important functions are described below.

Button	Explanation
	Opens the menu for configuring the advanced data recorder
	Removing elements from the current selection
	Saving the current selection to a file
	Opening a selection from a file
	Loading a predefined template
	Button for configuring a Start Trigger
	Button for configuring a Stop Trigger
 	Button for starting and stopping the recording
	Button for adding the desired variables to a Plot screen

The dialog “Confirm Merge Action” will appear and asking to merge or replace the items if there are already SysIDs selected and the submenu item “**Import Items**” is used to add a new or another template.



After clicking the “**Start Recording**” button the storage location and file name for the recording must be selected. Afterwards the data is recorded and a “Recording” status indicates an ongoing recording.



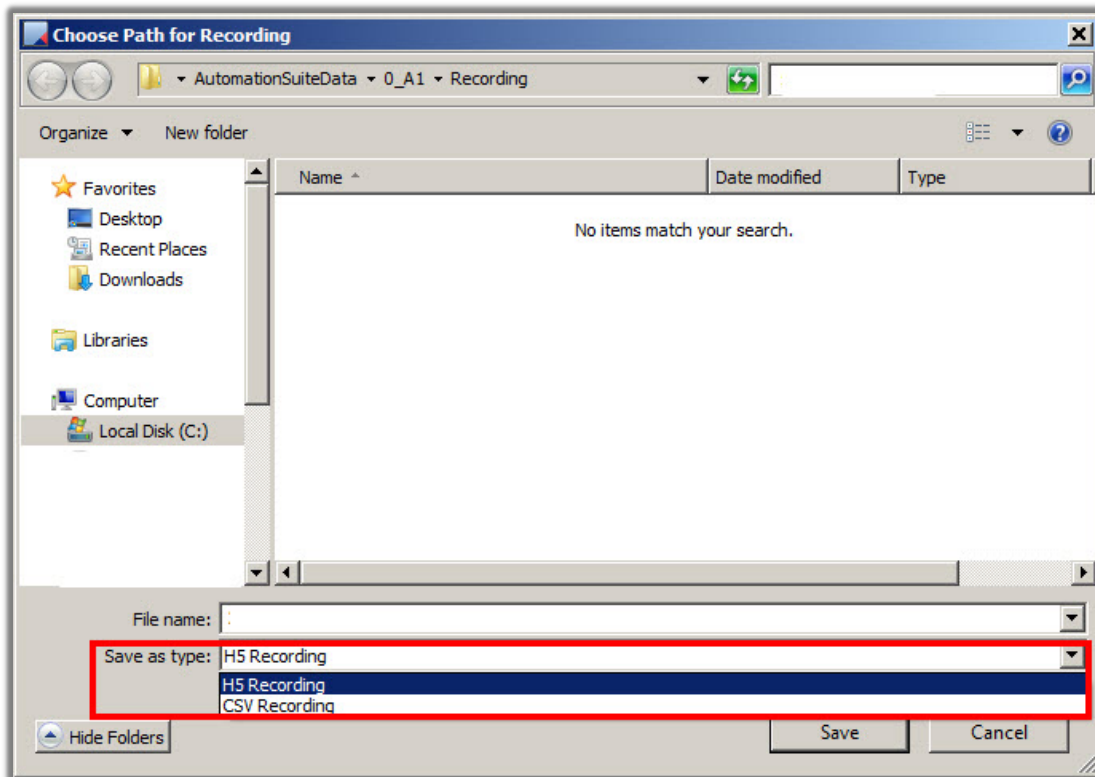
Note: Closing the configuration screen stops the recording.

14.3 Save or Export

By default the recording file has the format HDF.

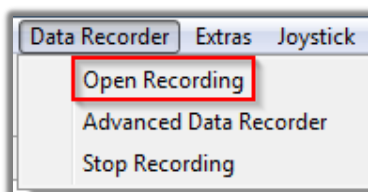
In the dialog “Choose Path for Recording” which appears before the recording is started the file format CSV can be selected.

Note: The CSV format cannot be opened in the data recorder for evaluation!



14.4 Evaluation

The recorded HDF file can be opened in the Automation Suite by using the submenu item “**Open Recording**” in menu “**Data recorder**” menu.



After the selected file has been opened it can be configured which measurement data (SysIDs) should be displayed. Of course, only those SysIDs are available, which were in the configuration of the recording selected.

Confirming the configuration will open a Plot screen with a timeline of the measured values (see chapter 12.5 *Plot screen*).

15 Configurator

The configurator can be started in the menu “**Configurator**” or via the **toolbar** icon “**Configurator**”.

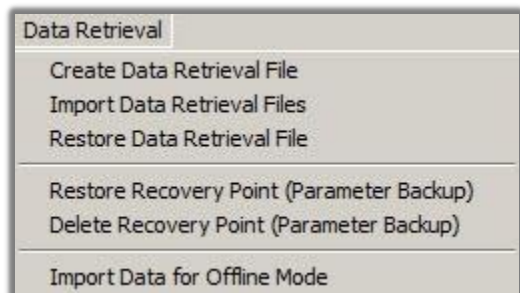


Note: The configurator functionality is not described in this document.

16 Data Retrieval

Menu item “**Data Retrieval**”.

The function “Data Retrieval” allows to save or restore the configuration of a SPU node.

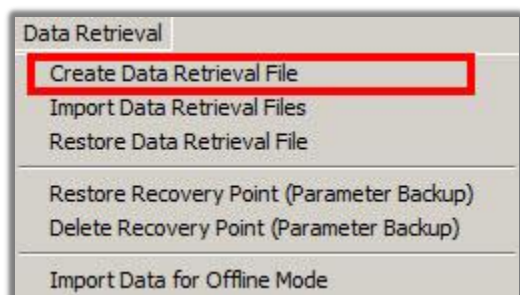


16.1 Create Data Retrieval File

Submenu item “**Create Data Retrieval File**” in menu “**Data Retrieval**” or via **toolbar** icon “**Create Data Retrieval File**” is used to create a data retrieval file of all available SPU nodes. This file can be stored in a directory defined by the user.

Depending on SPU1 or SPU2 being connected, one of two Yard Interface Settings dialogs will appear during the process and require interaction to store the current settings.

The function “Create Data Retrieval File” is not available in Offline-Mode.



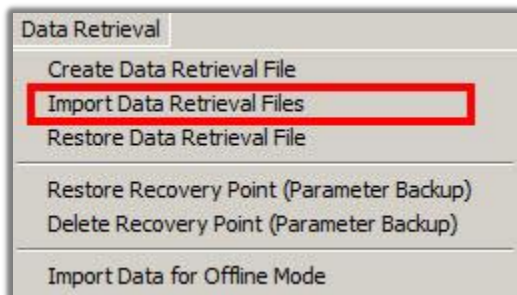
Shipyards Dialog for SPU1 (left) and SPU2 (right)



16.2 Import Data Retrieval Files

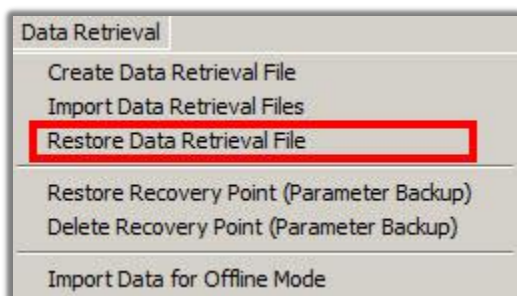
Submenu item **“Import Data Retrieval Files”** in menu **“Data Retrieval”** is used to copy all data retrieval files which are stored on the SPU SD-card to the service-laptop. The files can be stored in a directory defined by the user.

The function **“Import Data Retrieval Files”** is not available in Offline-Mode.

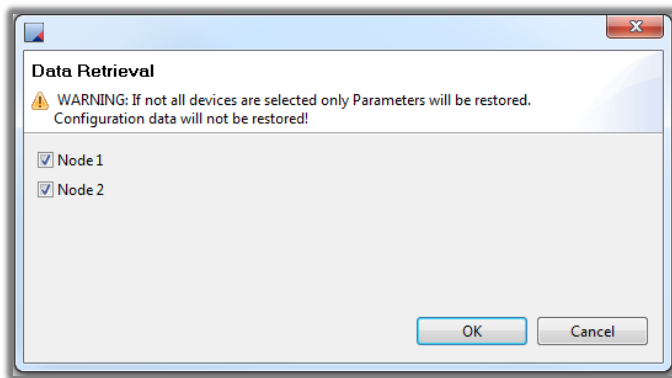


16.3 Restore Data Retrieval File

Submenu item **“Restore Data Retrieval File”** in menu **“Data Retrieval”** is used to restore the SPU configuration via data retrieval file. After selecting the data retrieval file the SPU nodes must be selected of which the data should be restored.



The setting of the Configurator is only restored, if all SPU nodes are selected. Otherwise only parameters (dialog) will be restored.



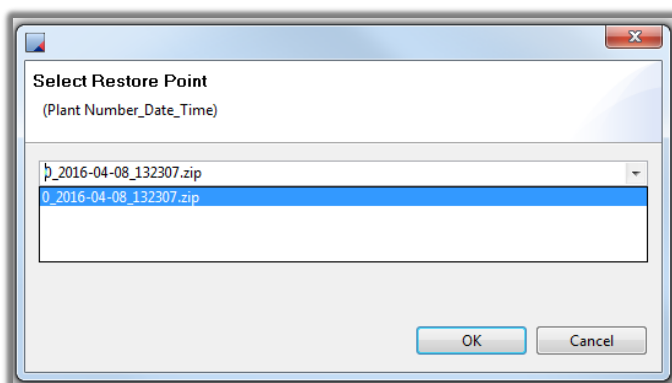
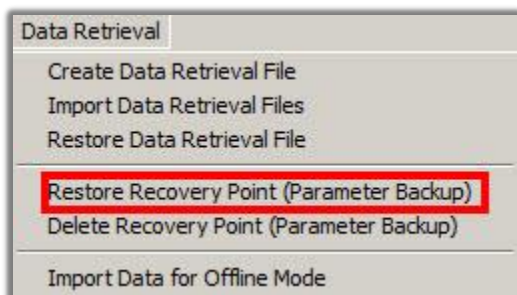
Note: It is not possible to restore a data retrieval file of a software with different MTU order number.

If the installed SPU software is not compatible with the data retrieval file, an upgrade function is started automatically.

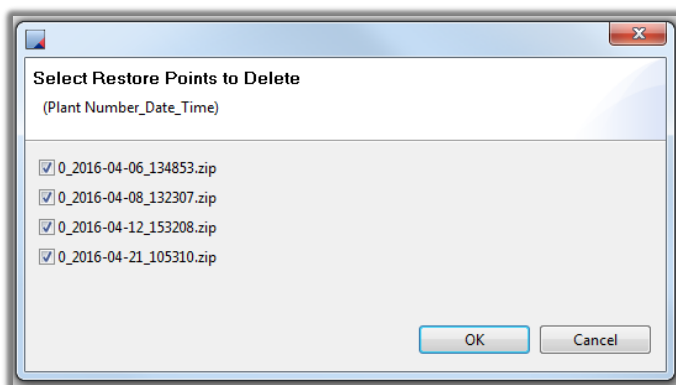
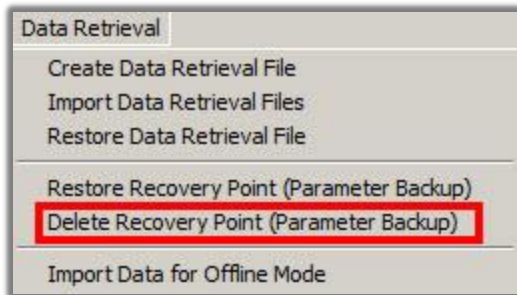
16.4 Recovery Point (Parameter Backup)

Submenu item “**Restore Recovery Point (Parameter Backup)**” in menu “**Data Retrieval**” is used to restore an automatically created recovery point. After selecting the recovery point the SPU nodes must be selected of which the data should be restored.

Note: An automatic parameter backup is created at the first parameter transfer each session.



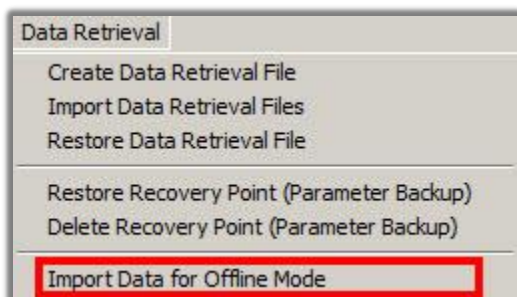
Submenu item “**Delete Recovery Point (Parameter Backup)**” in menu “**Data Retrieval**” is used to delete recovery points.



16.1 Import Data for Offline-Mode

Submenu item “**Import Data for Offline-Mode**” in menu “**Data Retrieval**” is used to copy required software files from the device to the service-laptop. The SPU nodes and the directory for the software files can be selected by the user. This function can be used to work in Offline-Mode (see chapter 9.2 *Offline-Mode*).

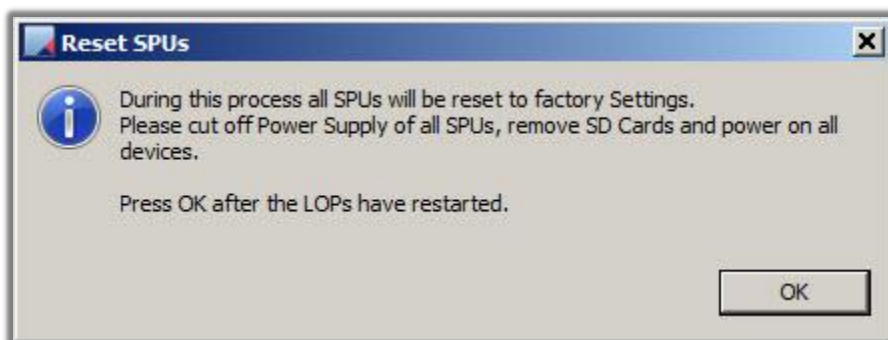
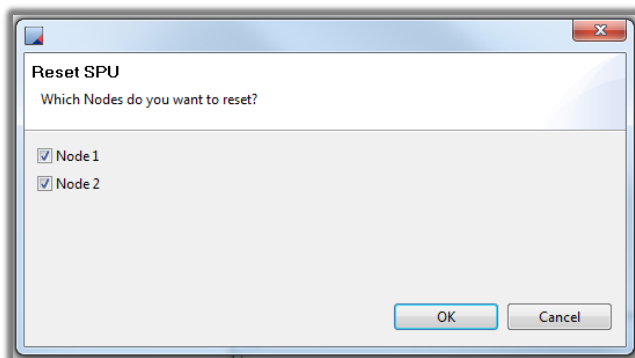
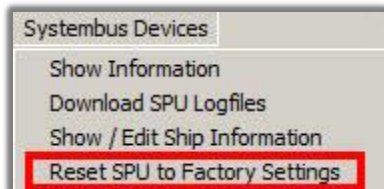
The function “Import Data for Offline-Mode” is not available in Offline-Mode.



17 Factory Reset

Submenu item “**Reset SPU to Factory Settings**” in menu “**Systembus Devices**” is reset the SPU to the delivery status. The software, the configuration and logfiles will be deleted.

This function should be used if a “wrong” software (e.g. other MTU order number) was installed.



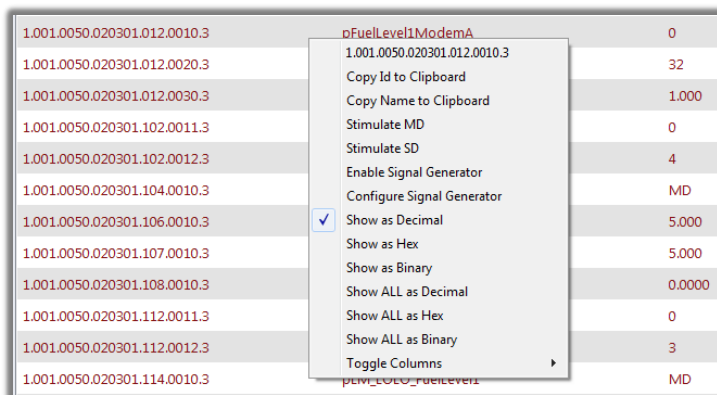
18 General Functions

18.1 Context menus

18.1.1 Context menu in the tabular view

A context menu that offers the following functions can be opened in the tabular view by right-clicking on the required column:

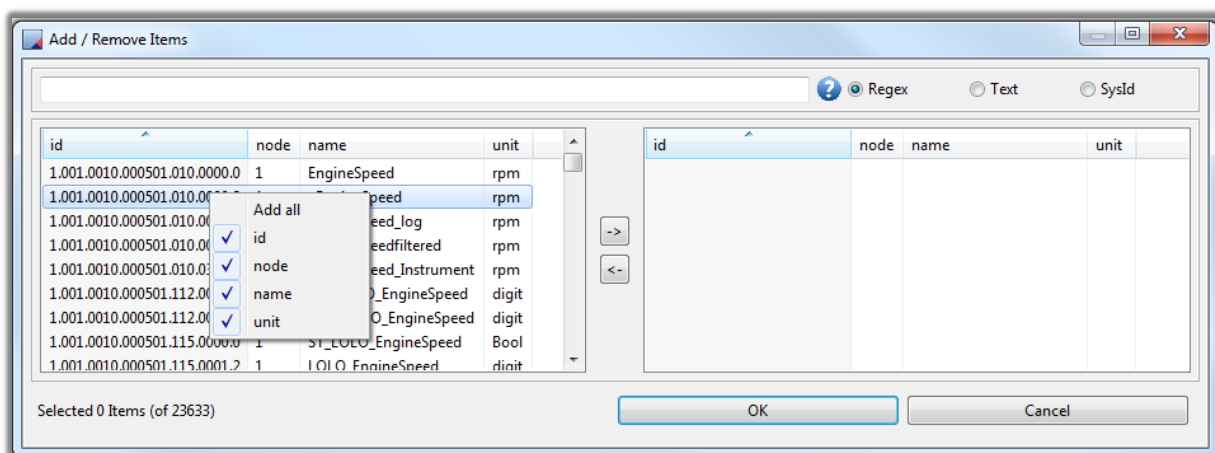
- Copy SysId number into the clipboard
- Copy SysId name into the clipboard
- Show SysId value in decimal notation (scaled)
- Show SysId value in hexadecimal notation (unscaled)
- Show SysId value in binary notation (unscaled)
- Show All SysId values from the current tabular view in decimal notation (scaled)
- Show All SysId values from the current tabular view in hexadecimal notation (unscaled)
- Show All SysId values from the current tabular view in binary notation (unscaled)
- Show/hide columns
- Stimulate SysId with value MD
- Stimulate SysId with value SD



Context menu in the tabular view

18.1.2 Context menu in selection dialog box

In a selection dialog box for SysIDs all values can be selected and deselected at once via the context menu. In addition it is possible to show and hide individual columns. The functions can be selected using the right mouse button.



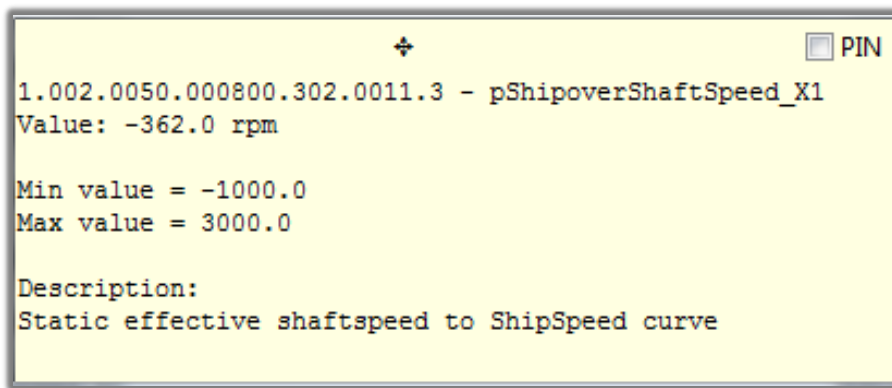
Selection dialog box context menu

18.2 Tooltip (SysId)

A tooltip is shown when the mouse is hovered over a SysId. The tooltip can be pinned by using the “**PIN**” Checkbox in the right upper corner. Then the tooltip is kept active even after the cursor is moved away.

In the tooltip the following **SysId information** are provided:

- Designation
- Number
- Current value
- Minimum value range
- Maximum value range
- Description



Tooltip using the example of a SysId

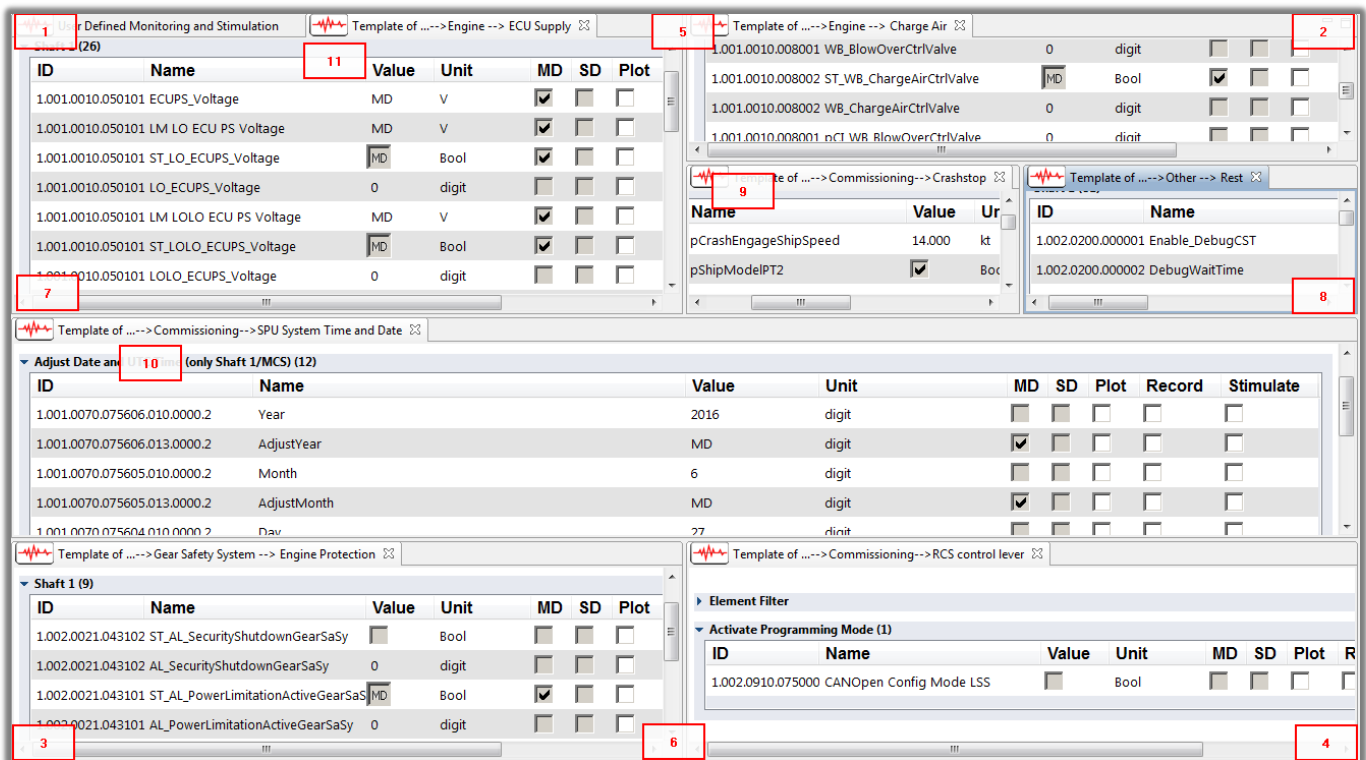
Note: The current value of the SysId is displayed without delay.
In the tabular views, the change rate is set lower to maintain legibility.

18.3 Window arrangement

The open windows can be moved and rearranged in most cases. The system supports the user through snapping and docking.

Independent of the number of open screens the snap points are at the following locations:

- 1-4: Corners of the Automation Suite
- 5-8: At the four outer edges
- 9-10: Horizontally or vertically between the already existing screens
- 11: As a new tab within an existing screen



The screenshot displays the BlueVision NewGeneration Automation Suite interface with several windows docked. Numbered red boxes indicate snap points for window arrangement:

- 1**: Top-left corner of the main workspace.
- 2**: Top-right corner of the main workspace.
- 3**: Bottom-left corner of the main workspace.
- 4**: Bottom-right corner of the main workspace.
- 5**: Top edge of the main workspace.
- 6**: Bottom edge of the main workspace.
- 7**: Left edge of the main workspace.
- 8**: Right edge of the main workspace.
- 9**: Horizontal snap point between the 'Defined Monitoring and Stimulation' window and the 'Template of ...->Commissioning-->Crashstop' window.
- 10**: Vertical snap point between the 'Adjust Date and Time' window and the 'Template of ...->Commissioning-->RCS control lever' window.
- 11**: A new tab within the 'Defined Monitoring and Stimulation' window.

The interface includes several data tables:

ID	Name	Value	Unit	MD	SD	Plot
1.001.0010.050101	ECUPS_Voltage	MD	V	☒	☐	☐
1.001.0010.050101	LM LO ECU PS Voltage	MD	V	☒	☐	☐
1.001.0010.050101	ST_LO_ECUPS_Voltage	MD	Bool	☒	☐	☐
1.001.0010.050101	LO_ECUPS_Voltage	0	digit	☐	☐	☐
1.001.0010.050101	LM LOLO ECU PS Voltage	MD	V	☒	☐	☐
1.001.0010.050101	ST_LOLO_ECUPS_Voltage	MD	Bool	☒	☐	☐
1.001.0010.050101	LOLO_ECUPS_Voltage	0	digit	☐	☐	☐

ID	Name	Value	Unit	MD	SD	Plot	Record	Stimulate
1.001.0070.075606.010.0000.2	Year	2016	digit	☐	☐	☐	☐	☐
1.001.0070.075606.013.0000.2	AdjustYear	MD	digit	☒	☐	☐	☐	☐
1.001.0070.075605.010.0000.2	Month	6	digit	☐	☐	☐	☐	☐
1.001.0070.075605.013.0000.2	AdjustMonth	MD	digit	☒	☐	☐	☐	☐
1.001.0070.075604.010.0000.2	Day	27	digit	☐	☐	☐	☐	☐

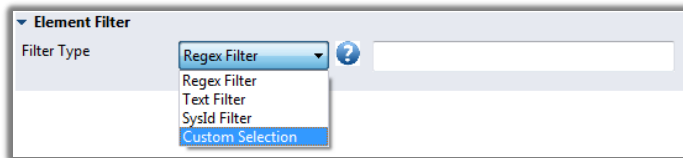
ID	Name	Value	Unit	MD	SD	Plot
1.002.0021.043102	ST_AL_SecurityShutdownGearSaSy	☐	Bool	☐	☐	☐
1.002.0021.043102	AL_SecurityShutdownGearSaSy	0	digit	☐	☐	☐
1.002.0021.043101	ST_AL_PowerLimitationActiveGearSaSy	MD	Bool	☒	☐	☐
1.002.0021.043101	AL_PowerLimitationActiveGearSaSy	0	digit	☐	☐	☐

ID	Name	Value	Unit	MD	SD	Plot	R
1.002.0910.075000	CANOpen Config Mode LSS	☐	Bool	☐	☐	☐	☐

18.4 Text filter

18.4.1 Text filters in the tabular view

There are **filter functions** available for each tabular view:



Selection of text filter in the tabular view

1. Text filter

When a string is entered a filtering for matching entries is automatically executed (every SysId information is included in filtering).

2. RegEx Filter (default)

Like the text filter, however, additional regular expressions can be used during input.

Other expressions that can be used:

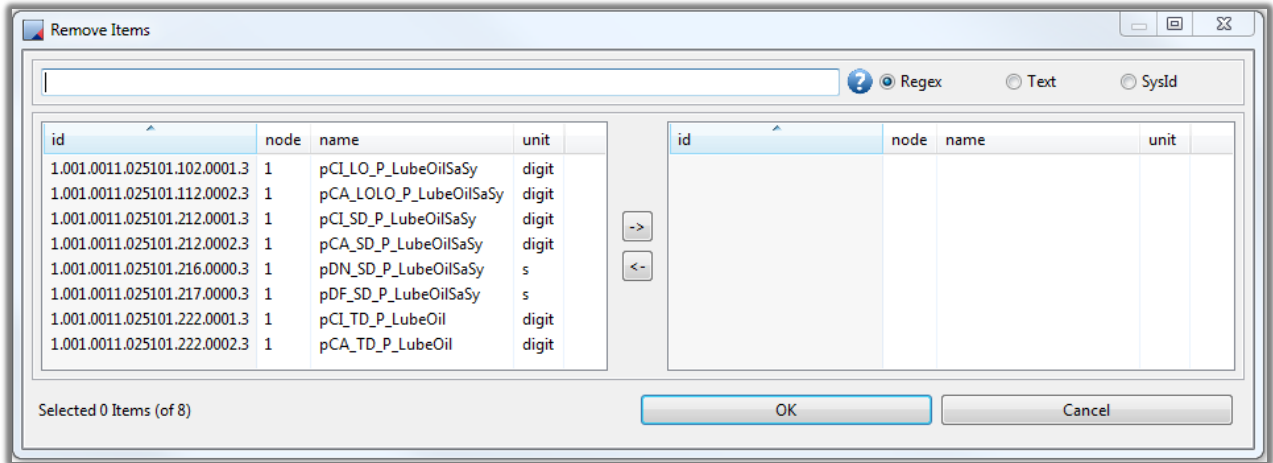
Printout	Example	Description
shaft	shaft=1	Filter for SysIDs from shaft 1
node	node=1	Filter for SysIDs from node 1
id	id=0010.000501.112.0001.3	Filter for SysIDs with number 0010.000501.112.0001.3
name	name=pCI_LOLO_EngineSpeed	Filter for SysIDs with name pCI_LOLO_EngineSpeed
unit	unit=digit	Filter for SysIDs with unit digit
type	type=param	Filter for SysIDs of the type Parameter
-	shaft=1, unit=digit	Filter for SysIDs with unit digit from shaft 1

3. SysId Filter

Corresponds to the text filter but when entering a SysId number non-existing “zeros” are ignored. For example, when entering “1.1.20.708.10.10.2”, the SysId “1.001.0020.000708.010.0010.2” is filtered anyhow.

4. Custom Selection

After selection of the filter type “**Custom Selection**”, two further buttons appear to allow elements to be removed or re-added in the active list. After clicking a button, the relevant selection dialog in which the previously described filters can be used appears.



Selection dialog for Custom Selection

The Custom Selection filter should be used to adapt and change predefined lists.

18.4.2 Text filter in the selection dialog box

Three filter functions are available in the selection dialog box:

1. Text filter

Behaves as in the tabular view (see chapter 18.4.1 *Text filters in the tabular view*).

2. RegEx Filter (default)

Behaves as in the tabular view (see chapter 18.4.1 *Text filters in the tabular view*).

3. SysId

Behaves as in the tabular view (see chapter 18.4.1 *Text filters in the tabular view*).

19 Directory Structure

When the Automation Suite is installed, a directory structure is created automatically under “C:\AutomationSuiteData”. After installation, the path can be changed under *Extras -> Preferences -> General*. The data to be saved are stored in this directory. The structure is described below.



1. For each connected plant data is stored in its own subdirectory.
 - a. Directory to store configurator values during each configuration process.
 - b. Default directory to store data retrieval files.
 - c. Default directory to store Templates in “Dialog”
 - d. Directory to store automatic recovery points.
 - e. Default directory to store Logfiles
 - f. Default directory to store SysId recordings
 - g. Default directory to store Templates in “Monitoring and Stimulation”
2. Default Automation Suite licence directory
3. Directory for general templates (views.zip used for all connected devices)
4. Internal directory for programs used during the update process

20 Troubleshooting

20.1 The password for the licence file has been forgotten

See chapter 7.7 *New Licence Password*

20.2 The licence file does not exist anymore

Send an e-mail to licence_request@mtu-online.com to retrieve the licence file.

20.3 The licence file was not detected

Restart the application.

Check if the licence file format is correct (.lic)

Check whether the licence path is correct (see chapter 7.6 *Licence Directory*).

Check whether a licence file was found (see chapter 7.5 *Switch Licence*).

20.4 No connection to plant/system possible

Check if the network cable is correctly connected to the systembus.

Check if the account number of the SPU has been set correctly.

Check if the SPU has loaded your project software (service display).

Check whether the correct IP settings are active on the Service Laptop (see chapter 4 *Preparation*).

20.5 SysIDs in the tabular views are not displayed

Check if the correct view has been selected (only parameters are visible in the dialog).

You possibly do not have the required permission.

20.6 No change of SysIDs (parameter or stimulation) possible

You possibly do not have the required permission.

20.7 Screen flickering or black screen

See chapter 4.4 *Visual Effects in Windows 7*

20.8 Connectivity problems with external network card

Try to use your computer's internal network card.